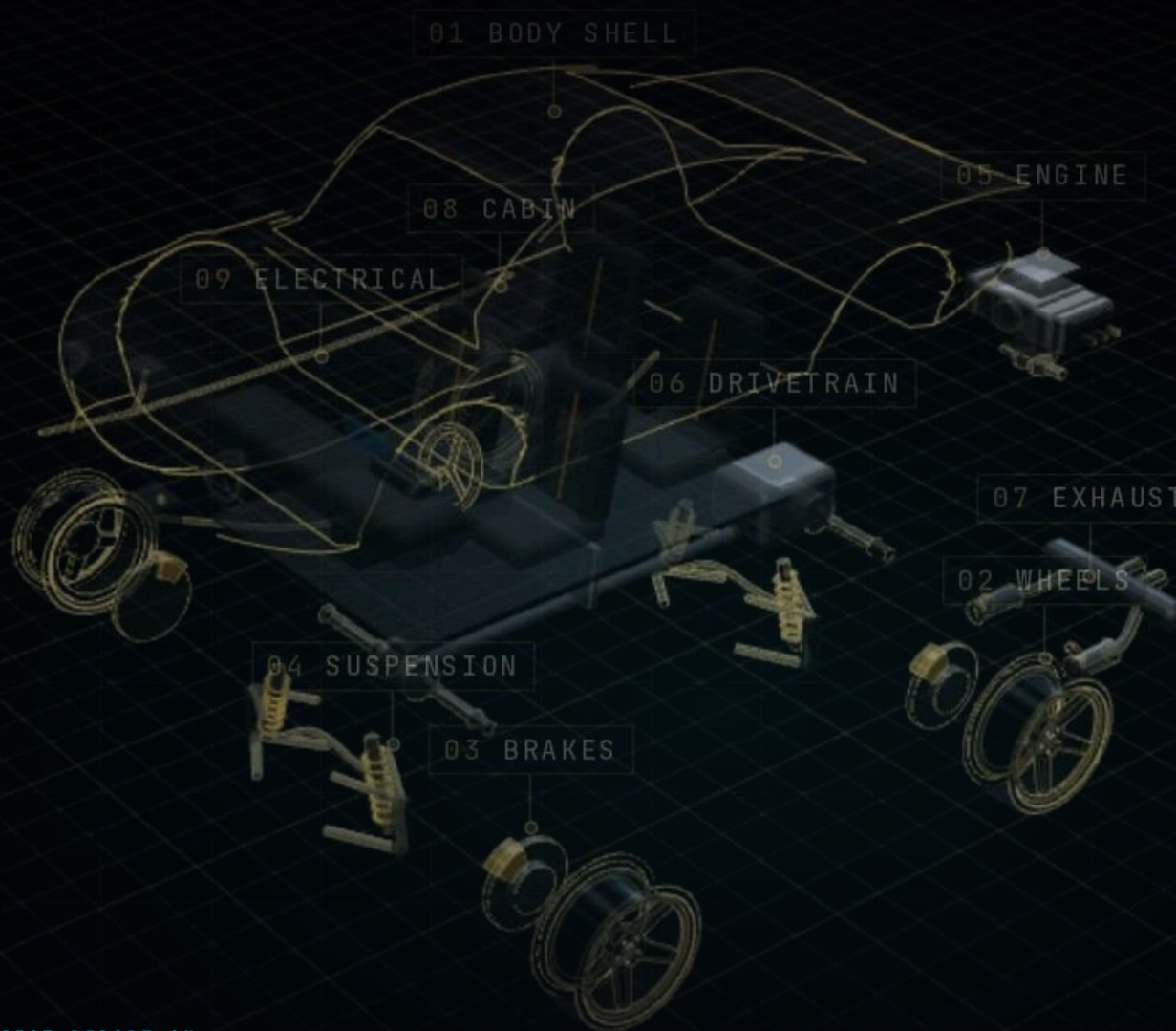




A MINI PROJECT REPORT ON

AURIX

GLOBAL AUTOMOTIVE INTELLIGENCE

A premium 3D interactive automotive encyclopedia, visualizer and configurator platform.

SUBMITTED BY

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UNDER THE GUIDANCE OF

Amrut Nikam

ACADEMIC YEAR

2026 - 27

INDEX

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Structure follows the department's mini project report format. Every screenshot in Section 09 is captured from the running application.

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Every figure published.
Every gap marked.

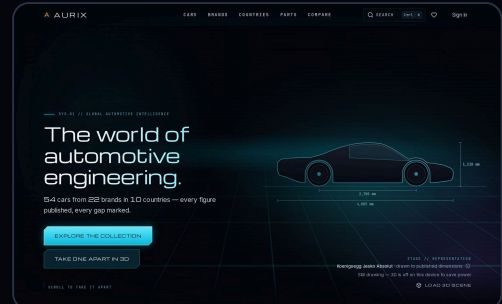
AURIX is a full-stack web platform that brings global automotive data, interactive 3D and structured engineering information together in one place.

Visitors move through a clear hierarchy, from **country** to **manufacturer**, **model**, **variant**, **specifications** and individual **parts**. Each car has its own page. It has a 3D viewer, an engineering tour driven by scrolling, a four-view blueprint drawn to published dimensions, a feature showcase, full technical data and sourced prices by market.

A plain-English search ("german supercars above 300 km/h"), shareable filters, a side-by-side comparison of up to four cars, favourites and a complete admin CMS finish the product.

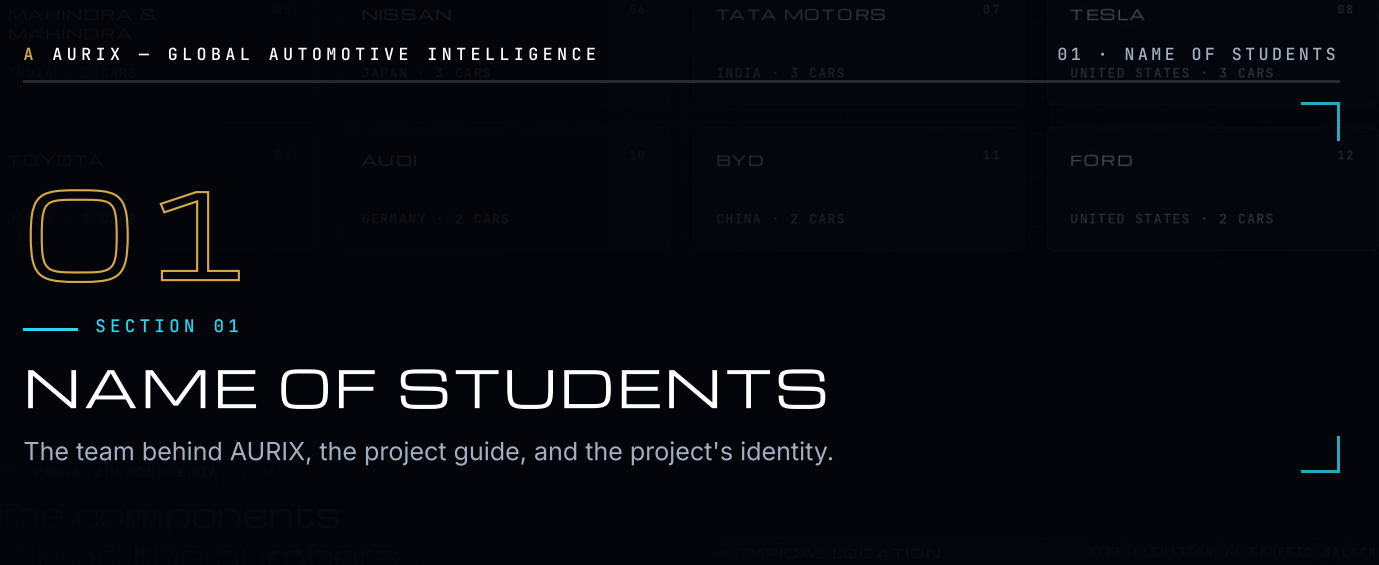
The platform is built with **Next.js 16**, **React 19**, **TypeScript**, **React Three Fiber** and **Supabase (PostgreSQL)**. Row Level Security protects every table.

One rule runs through the whole project: **no figure is ever invented**. If a maker does not publish a number, it stays empty in the database and the site shows "Not available".



An infographic with a dark blue background and a light blue grid. It displays eight metrics in a 2x4 grid. Each metric is represented by a large white number and a smaller white label below it. Horizontal lines are placed above and below each row of metrics.

10	22	46	54
COUNTRIES	MANUFACTURERS	CAR MODELS	VARIANTS
71	27	858	10
PARTS DOCUMENTED	DATABASE TABLES	AUTOMATED TESTS	3D SUBSYSTEMS



"AURIX — Global Automotive Intelligence" was designed and developed as a group project by the following students, in partial fulfilment of the requirements of the B.Tech Computer Science and Engineering programme.

1.1 Project team

S.NO.	NAME OF STUDENT	ROLL NO. / ENROLLMENT NO.	CLASS / SEMESTER
01	Zenil Rana	39	B.Tech CSE
02	Sairaj Bhadale	6	B.Tech CSE
03	Saket Shrivastava	48	B.Tech CSE

1.2 Guide / mentor

GUIDE NAME	ROLE	DEPARTMENT
Amrut Nikam	Project Guide (Faculty)	B.Tech CSE

1.3 Project identity

ITEM	DETAIL
Title	AURIX — Global Automotive Intelligence
Type	Group mini project · full-stack web application
Domain	Web development · 3D web graphics · relational databases
Institution	Pimpri Chinchwad University · Department of B.Tech CSE
Academic year	2026 – 27

02

SECTION 02

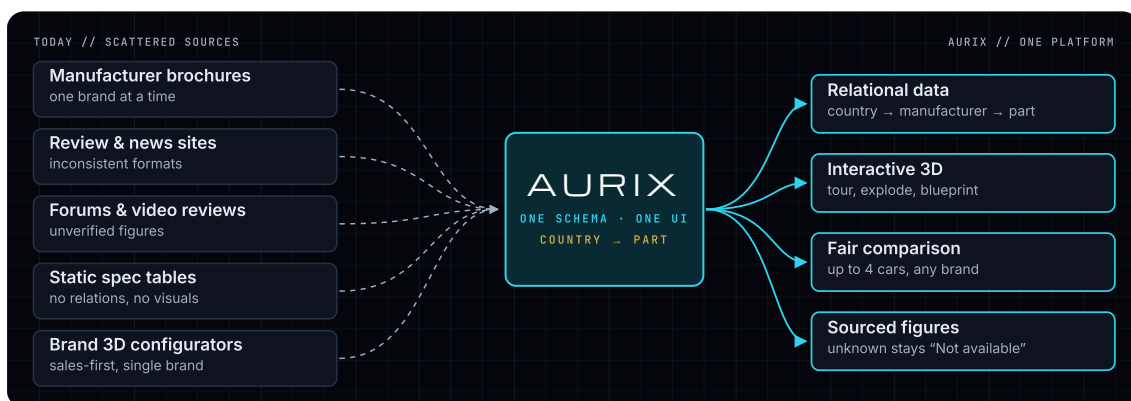
PROBLEM STATEMENT

Automotive information is everywhere, but it is scattered, inconsistent and rarely explorable.

Enthusiasts, students and buyers find automotive information in dozens of places that do not connect. Brochures, review sites, forums, video reviews and static specification tables all use different formats. They rarely link related facts, such as how a country's industry shaped a manufacturer, or how a model grew into its variants. Almost none let a user explore how a car is built, visually and interactively.

Premium brands do offer 3D configurators, but each covers one maker's own cars and exists to sell. None of them compares vehicles across brands, countries or powertrain types. So there is no neutral, technically rich platform that combines:

- Country-wise organisation of manufacturers and their history.
- A relational database linking countries, manufacturers, models, variants and individual parts.
- Interactive 3D, including an exploded component view and an engineering visualization.
- Consistent specification data (performance, engine, transmission, dimensions, fuel or energy) for every powertrain.
- Advanced search, filtering and side-by-side comparison of several vehicles.



STATEMENT OF THE PROBLEM

"To design and develop a unified, database-driven, 3D-interactive web platform that presents global automotive information – spanning countries, manufacturers, models, variants, specifications and individual vehicle parts – in a structured, comparable and visually engaging manner, while remaining technically extensible for future data growth and administrative management."

03

SECTION 03

ALL-WHEEL DRIVE

PROJECT DESCRIPTION

A 3D configurator, a global encyclopedia, an engineering visualizer and a comparison tool in one product.

3.1 Overview of the project

AURIX is a full-stack, database-backed web application. It works as a 3D automotive viewer, a global automotive encyclopedia, an engineering visualization platform and a car comparison tool. The frontend uses **Next.js 16** and **React 19**, with **React Three Fiber / three.js** for real-time 3D and **Tailwind CSS v4** for a futuristic automotive HUD design language. **Supabase (PostgreSQL)** is the relational backend, with authentication and storage.

All data follows one hierarchy. A user can start broad, with a country or a manufacturer, and drill down to precise mechanical detail: the torque of one engine variant, or where the turbochargers sit in a specific car.



3.2 Objectives

OBJECTIVE	STATUS
A normalised relational schema for countries, manufacturers, models, variants, specifications and parts, without duplication.	● DELIVERED
An interactive 3D viewer with orbit, zoom, pan and preset camera views.	● DELIVERED
An exploded view that separates a car into its major assemblies, each with an information panel.	● DELIVERED
Country-wise and manufacturer-wise browsing with global search and multi-parameter filters.	● DELIVERED
Side-by-side comparison of two to four vehicles.	● DELIVERED
A responsive, accessible and fast platform on desktop and mobile.	● DELIVERED
An administrative content-management layer.	● DELIVERED · FULL CMS

3.3 Scope of the project

The finished scope covers the full system architecture, the relational schema (**27 tables** and **2 views** across 8 migrations), the 3D pipeline, and every user-facing module: home, catalogue, car detail, manufacturers, countries, parts, search, comparison, favourites and accounts. It also includes a complete **admin CMS** for vehicles, models, prices, markets, media and data sources.

The seed data covers **10 countries, 22 manufacturers, 46 models, 54 variants and 71 documented parts**. The schema scales to thousands of vehicles without structural change.

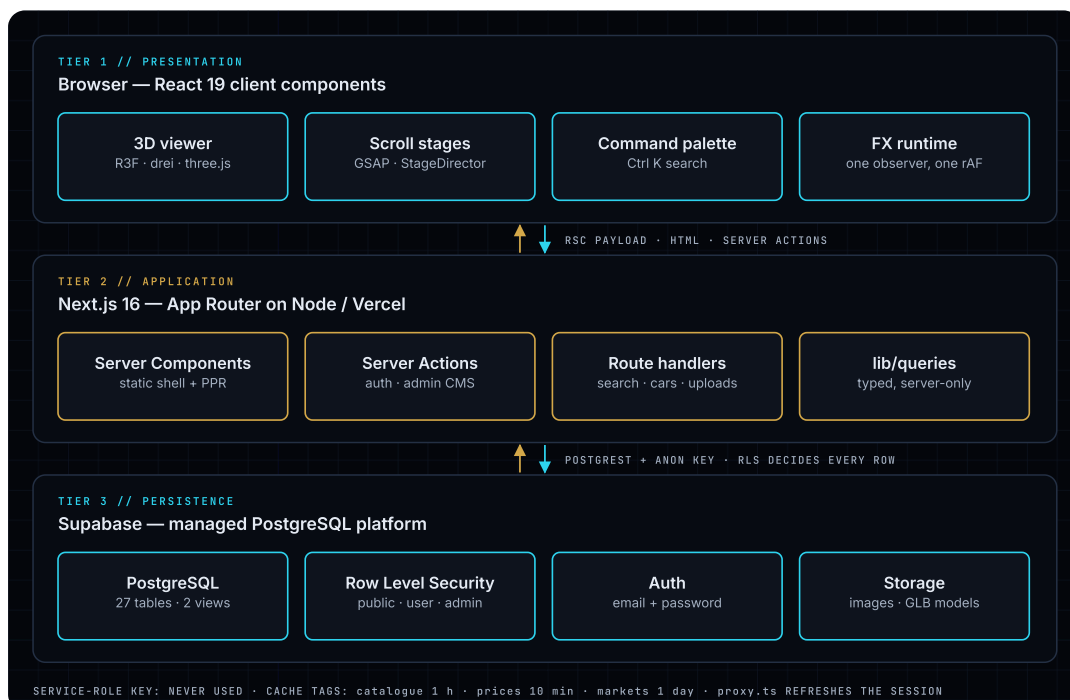
3.4 Key features and modules

MODULE	WHAT IT DOES
Cinematic homepage	Hero stage with the car drawn to published dimensions (3D on capable devices), a scroll-driven anatomy story, categories, origins, brands and the parts encyclopedia.
3D car viewer	React Three Fiber viewer: orbit, zoom and pan, nine camera presets, four lighting setups, hotspots and quality tiers chosen automatically.
Exploded view	Separates the car into up to ten subsystems (body, engine, transmission, suspension, brakes, wheels, interior, electronics, battery, exhaust), with a panel for each.
Engineering tour	A scroll-driven tour through the car's own layout: engine or motors, battery, drivetrain, chassis, brakes and cabin, showing only the figures that exist.
Blueprint	A four-view drawing (side, front, top, rear) at one scale, with dimension lines only for published figures, plus a 3D blueprint chapter.
Car detail page	Identity, gallery, performance, Car DNA, powertrain diagram, design, features, technical data and pricing. Sections follow the fuel type, so an EV never shows an engine.
Anatomy of the machine	A parts encyclopedia: 71 parts in 9 systems, each linked to the cars that use it.

MODULE	WHAT IT DOES
Country explorer	An interactive world map and country pages with automotive history, manufacturers and models.
Manufacturer pages	Brand profile (origin, founding year, headquarters, story) and a model grid.
Variant system	Several variants per model (e.g. M3, M3 Competition, M3 Competition xDrive), each with its own specifications.
Search and filters	Command palette (Ctrl K) with a rule-based plain-English parser, plus URL-driven filters for country, brand, body, fuel, drive, transmission, power, top speed and year.
Comparison tool	Up to four cars in one table. A “best” marker appears only when at least two cars publish that figure. Prices are never converted or ranked.
Sourced pricing	Prices by country, state and city with price type, source, source URL and verification date. On-road totals are calculated only when every component is known.
Car DNA and powertrain	A percentile profile computed from real figures, and an animated energy-flow diagram for petrol, diesel, electric and hybrid drivetrains.
Accounts and favourites	Email sign-in, favourites for guests (on the device) and accounts (in the database), merged at sign-in, and a recently-viewed history.
Admin CMS	Role-guarded dashboard to edit every entity, record prices with sources, import prices from CSV, upload validated media and track data provenance.

3.5 System architecture

The application uses a three-tier architecture. UI components never query the database directly: they call typed, server-only functions in `src/lib/queries`, and the database decides every row through Row Level Security.



3.6 Database design

The database is a normalised PostgreSQL schema that mirrors the hierarchy. The four specification tables are one-to-one with a variant, so each figure has exactly one home. Where a value is not published, the column stays NULL.

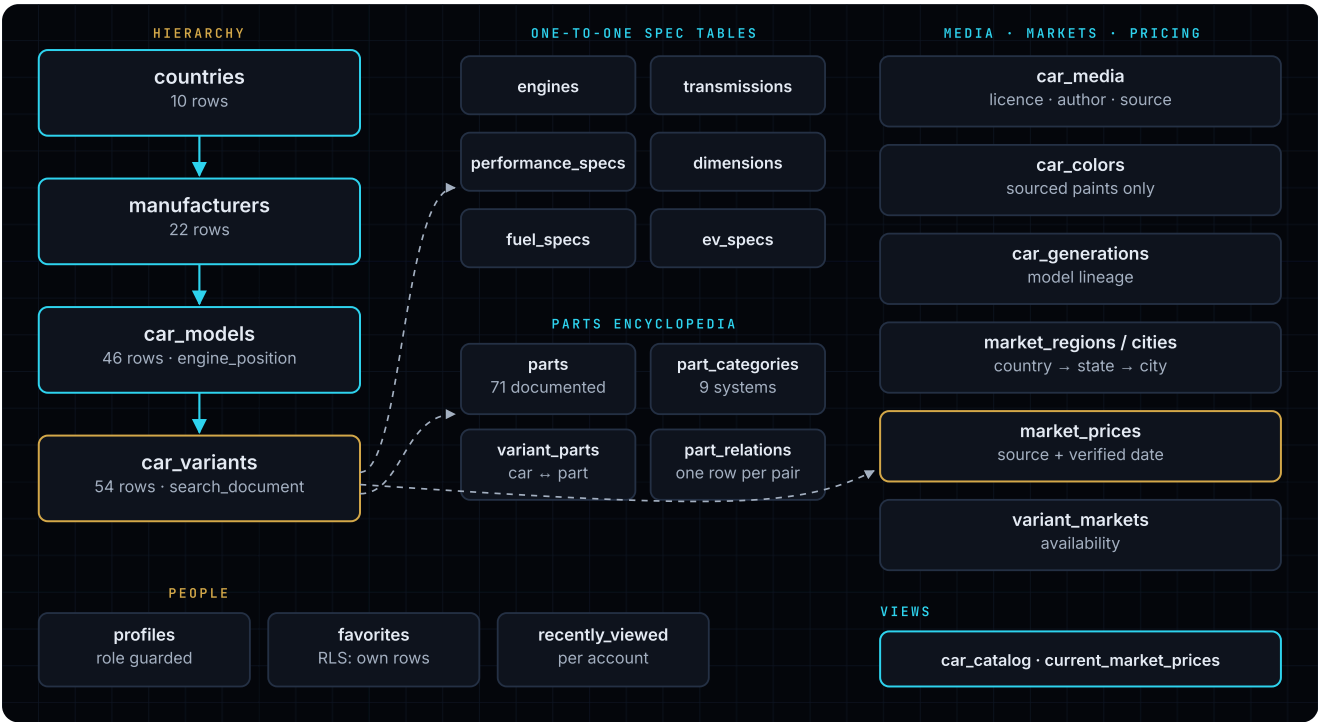


TABLE GROUP	PURPOSE
countries · manufacturers · car_models · car_variants	The browsing hierarchy. Variants carry a full-text search_document kept current by a trigger.
engines · transmissions · performance_specs · dimensions · fuel_specs · ev_specs	Specifications, each with source, source_url and last_verified_at. Triggers enforce the powertrain rules.
parts · part_categories · variant_parts · part_relations	The parts encyclopedia and its cross-references (one row per pair).
car_media · car_colors · car_generations · features · variant_features · categories	Photographs and 3D models with licence and author, sourced paint colours, lineage, features and categories.
market_regions · market_cities · market_prices · variant_markets	Markets and sourced prices. current_market_prices picks the latest price in force.
profiles · favorites · recently_viewed	Users. A trigger protects profiles.role, and RLS limits each user to their own rows.

3.7 3D visualization architecture

The 3D experience follows one pipeline. The car is not a fixed mesh: it is a **lofted parametric surface** built from a side profile for each body style and sized to the variant's published dimensions. Its insides are laid out from the variant's own rows: cylinder count and bank layout, engine position (front, mid or rear), driven axles, and motor count and battery for EVs. A 911 gets a flat-six behind the rear axle; a Model S Plaid gets three motors and a floor battery, and no engine.



A licensed **GLB model** (Draco, KTX2 and meshopt supported, self-hosted decoders) replaces the parametric car when one is uploaded, and is labelled "exact model" or "representation". The canvas mounts only when it scrolls near the viewport and pauses when it leaves. Lighting is built in the scene, so no third-party HDR is fetched at runtime.



FIG 3.1 3D blueprint: the body shell drawn as a ghost with gold panel lines.



FIG 3.2 Exploded finale: every system moved out along its own path.

3.8 Technology stack justification

Next.js 16 (App Router, Turbopack, Cache Components) serves an SEO-friendly, content-heavy catalogue. Every car, brand, country and part page is prerendered as a static shell, with small dynamic parts such as the account menu streamed in. **React Three Fiber** lets the 3D scene live in the same React tree as the UI. **GSAP ScrollTrigger** drives the scroll stages through refs rather than React state. **Supabase** provides managed PostgreSQL, authentication, storage and Row Level Security without a custom backend or Docker.

All versions are pinned exactly, so every build of the project is identical and reproducible:

PACKAGE	VERSION	ROLE
next	16.3.5	Framework: routing, rendering, server actions
react / react-dom	19.2.8	UI library (held below 19.3 for R3F 9 compatibility)
typescript	5.9.3	Static typing (held for typescript-eslint support)
three	0.186.0	WebGL 3D engine
@react-three/fiber · drei	9.7.0 · 10.7.8	Declarative 3D in React and helpers
gsap	3.15.0	Scroll-driven animation (ScrollTrigger)
@supabase/supabase-js · ssr	2.116.0 · 0.12.7	Database client and cookie-based auth
tailwindcss	4.3.3	CSS-first design tokens and utilities
lucide-react	1.47.0	Icon set
vitest · eslint	5.0.1 · 9.39.5	Unit tests and linting

Performance: initial JavaScript is about 190–240 KB gzipped per page. The ~262 KB three.js chunk is not in the first load of any page; it loads only when a viewer scrolls into view. Ambient animations pause off-screen, and slower devices switch automatically to a lighter "lite" mode.

3.9 Security architecture

Security is enforced at the **database**, not only in application code. Only the public URL and anonymous key reach the browser. **The service-role key is never used anywhere in the codebase.**

RISK	CONTROL IN AURIX
Reading unpublished or private data	Row Level Security on every table; views declared <code>security_invoker</code> so they cannot bypass RLS.
Promoting yourself to admin	A trigger restores <code>profiles.role</code> on every update, so a user cannot change their own role.
Forged sessions	<code>getUser()</code> revalidates the token with Supabase on every protected request; <code>getSession()</code> is never trusted.
Probing the admin area	<code>/admin</code> answers identically to “signed out” and “signed in without the admin role”.
Account enumeration	Sign-in failures return one generic message for both wrong email and wrong password.
Malicious uploads	Files are checked by magic bytes and headers (PNG, JPEG, WebP, AVIF sizes; glTF binary header and JSON chunk), not by name or declared type.
Bad data entry	Every server action validates its input; a price cannot be saved without source, source URL and verification date.

3.10 Advantages of the system

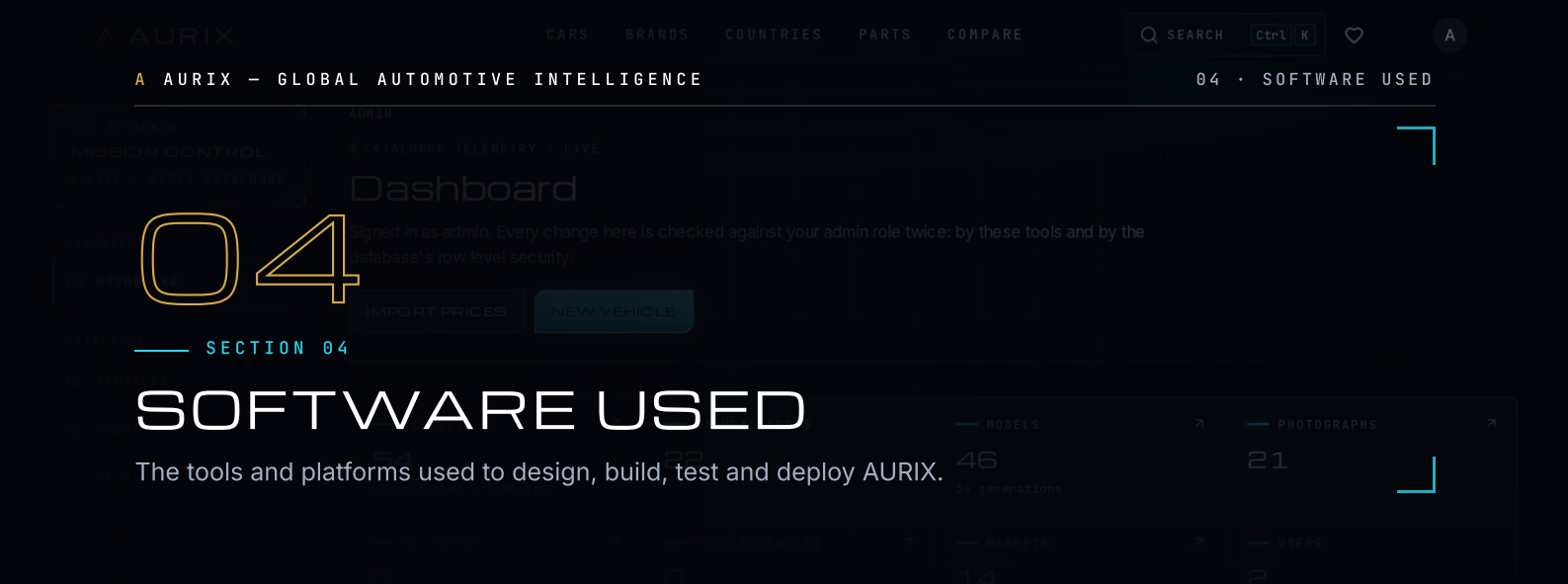
- Consolidates fragmented automotive information into one consistent, relational platform.
- Explains how cars are built through 3D, exploded views, a scroll tour and blueprints rather than static text.
- Brand-neutral, so cars from any maker and country can be compared fairly.
- Honest by design: every figure is published or marked “Not available”, and prices carry their source.
- Works without JavaScript for the catalogue and filters, and meets WCAG AA contrast on all body text.
- Scales from the seed data to thousands of vehicles without schema changes, with a CMS already in place.

3.11 Limitations and future scope

- The dataset is representative, not exhaustive: 54 variants today. Adding every global model is ongoing data entry through the CMS.
- No licensed GLB model is bundled, so every car currently shows the parametric representation. Exact models need licensed files.
- Prices, availability and paint colours are recorded only with a source. None are invented, so most cars show “Price data unavailable” until an admin records one.
- Some cars still use the silhouette placeholder until a verified, licensed photograph is added.
- On Vercel, admin uploads above 4.5 MB need signed direct-to-storage uploads.
- Future work: user reviews, multi-language support, more markets, and live price feeds where a licensed source exists.

3.12 Application route map

ROUTE	DESCRIPTION
/	Cinematic homepage with hero stage, anatomy story and entry points.
/cars	Catalogue with plain-English search, filters and sorting.
/cars/[maker]/[model]/[variant]	Car detail page, e.g. /cars/porsche/911/turbo-s.
/cars/[maker] • /cars/[maker]/[model]	Brand and model landing pages with variant selection.
/manufacturers • /manufacturers/[slug]	All brands, and a brand profile with its models.
/countries • /countries/[slug]	World map, and country history with its makers.
/parts • /parts/[slug]	Parts encyclopedia and individual part pages.
/compare	Side-by-side comparison of up to four cars (state kept in the URL).
/favorites • /account • /login	Saved cars, account settings and authentication.
/admin/*	CMS: vehicles, models, prices and CSV import, markets, media, sources.
/api/search • /api/cars • /api/favorites	Command palette search, card lookups and favourites.



CATEGORY	SOFTWARE / TOOL	PURPOSE
Code editor	Visual Studio Code	Writing and debugging the application code.
Runtime	Node.js 22 LTS	Runs the Next.js build and development server.
Framework	Next.js 16 (React 19)	Routing, rendering, server actions and route handlers.
Package manager	npm	Dependencies and scripts (dev, build, verify, db:push, db:seed, doctor).
Database platform	Supabase	Managed PostgreSQL, authentication, storage and RLS.
Database engine	PostgreSQL	The relational engine underneath Supabase.
Migrations	Supabase CLI	Applies the 8 SQL migrations directly (no Docker needed).
Version control	Git & GitHub	Source history, branches and pull requests.
Continuous integration	GitHub Actions	Lint, format check, typecheck, 858 unit tests and a build on every push.
Testing	Vitest · Playwright	Unit tests for the pure logic; browser checks at several screen widths.
Code quality	ESLint · Prettier	Static analysis and consistent formatting.
Design tool	Figma	UI/UX layout and visual planning.
3D asset tools	Blender / glTF utilities	Preparing optional licensed GLB models.
Browser tools	Chrome DevTools	Performance traces, responsive testing and debugging.
Deployment	Vercel	Hosting and continuous deployment.

4.1 Hardware and software requirements

The following configuration is recommended for development and for people using the deployed site in a browser.

REQUIREMENT	DEVELOPMENT MACHINE	END USER (BROWSER)
Processor	Intel i5 / AMD Ryzen 5 or higher	Any modern dual-core processor
RAM	8 GB minimum (16 GB recommended)	4 GB minimum
Graphics	Integrated GPU is enough; a dedicated GPU speeds up 3D preview	WebGL-capable GPU and driver
Operating system	Windows 10/11, macOS or Linux	Any OS with a modern browser
Runtime	Node.js 22.12 or newer	—
Browser	Latest Chrome, Edge or Firefox	Latest Chrome, Edge, Firefox or Safari
Internet	Required for Supabase and package installation	Required to load data and 3D

Adaptive by design: the site measures each device and adapts. Low-power devices show the still hero drawing and the lighter 3D recipe. Phones get a coarser car without shadow maps. Visitors who prefer reduced motion get every section at once, with nothing moving.

4.2 Commands used in development

COMMAND	PURPOSE
<code>npm run dev</code>	Development server at http://localhost:3000
<code>npm run verify</code>	Lint, typecheck, tests and production build: run before every release
<code>npm run db:push</code>	Apply the SQL migrations to the hosted database
<code>npm run db:seed</code>	Load the seed data (idempotent, safe to re-run)
<code>npm run doctor</code>	Check environment, DNS, key, migrations and seed in order

Dimensions & Weight

05

SECTION 05

• POWER CV	650 hp
• POWER AT	Not available
• TORQUE	800 Nm
• TORQUE AT	Not available

PROGRAMMING LANGUAGE USED

A strongly typed web stack: one set of types from the database to the screen.

LANGUAGE / TECHNOLOGY

ROLE IN THE PROJECT

TypeScript

The main language for components, data access, server actions and the 3D system. Strict mode, `noUncheckedIndexedAccess` and `no any`, so every optional figure is handled.

JavaScript (ES2022+)

What the browser and Node.js run; TypeScript compiles to it. Tooling scripts (seed, doctor, type generation, image fetching) are plain Node modules.

SQL (PostgreSQL • PL/pgSQL)

Schema, foreign keys, indexes, Row Level Security policies, triggers (search document, powertrain rules, role guard) and the search function.

HTML5

Semantic structure: one h1 per page, landmarks, labelled inputs, and JSON-LD structured data for cars and brands.

CSS3 / Tailwind CSS v4

The HUD design system: tokens in `@theme`, glass panels, chamfers, scanlines and animations that only use transform and opacity.

GLSL (via three.js / WebGL)

Shader programs behind the materials, lighting, reflections and shadows of the 3D viewer.

Types are generated from the live database schema (`src/types/database.ts`). The same types describe the data on the server and in the UI, so a mismatch between what the database returns and what a component expects fails at build time instead of in front of a user.

06

SECTION 06

APPLICATION AREA

An academic project whose architecture applies directly to real-world automotive products.

AURIX is presented here as a mini project, but its concept and architecture apply to several real-world domains:



01

Automotive education

Engineering and design students see how a car is laid out: engine position, driven axles, battery placement and how parts connect.



02

Digital automotive museums

Each country's automotive history and notable models, presented interactively.



03

Showroom kiosks

A touch-friendly 3D display that shows a car's real figures and components to walk-in customers.



04

Dealership listings

Sourced, market-specific prices by country, state and city, with clear price types.



05

Journalism and research

A structured reference for comparing specifications across makers and markets, with provenance.



06

Enthusiast communities

A brand-neutral space to explore, compare and share cars with a single link.



07

Manufacturer microsites

The viewer, tour and exploded view can be adapted into a single-brand configurator.



08

E-learning platforms

The powertrain visualizer teaches combustion, hybrid and electric energy flow.

07

SECTION 07

FLOWCHART

How the data is structured, and how visitors and administrators move through the system.

OVERVIEW

DESIGN

ENGINE

BRAKES

PERFORMANCE

7.1 Data hierarchy flowchart

Automotive data is structured relationally, from the broadest entity (country) to the most granular (individual parts). Every level is a one-to-many relationship enforced by foreign keys.

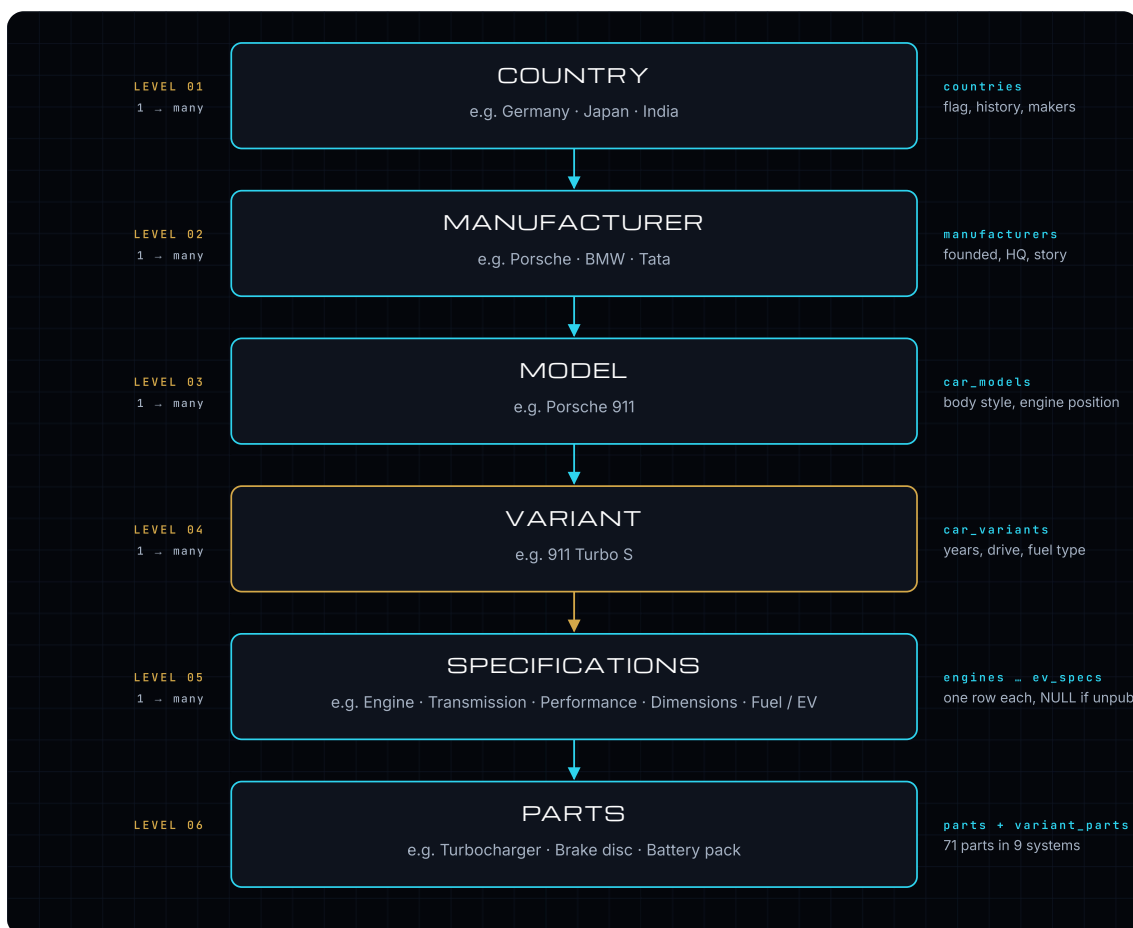


FIG 7.1 Data hierarchy: Country → Manufacturer → Model → Variant → Specifications → Parts

7.2 System process flowchart — visitor

A typical visit starts at the loading screen and branches through browsing, search, car exploration, comparison and favourites.

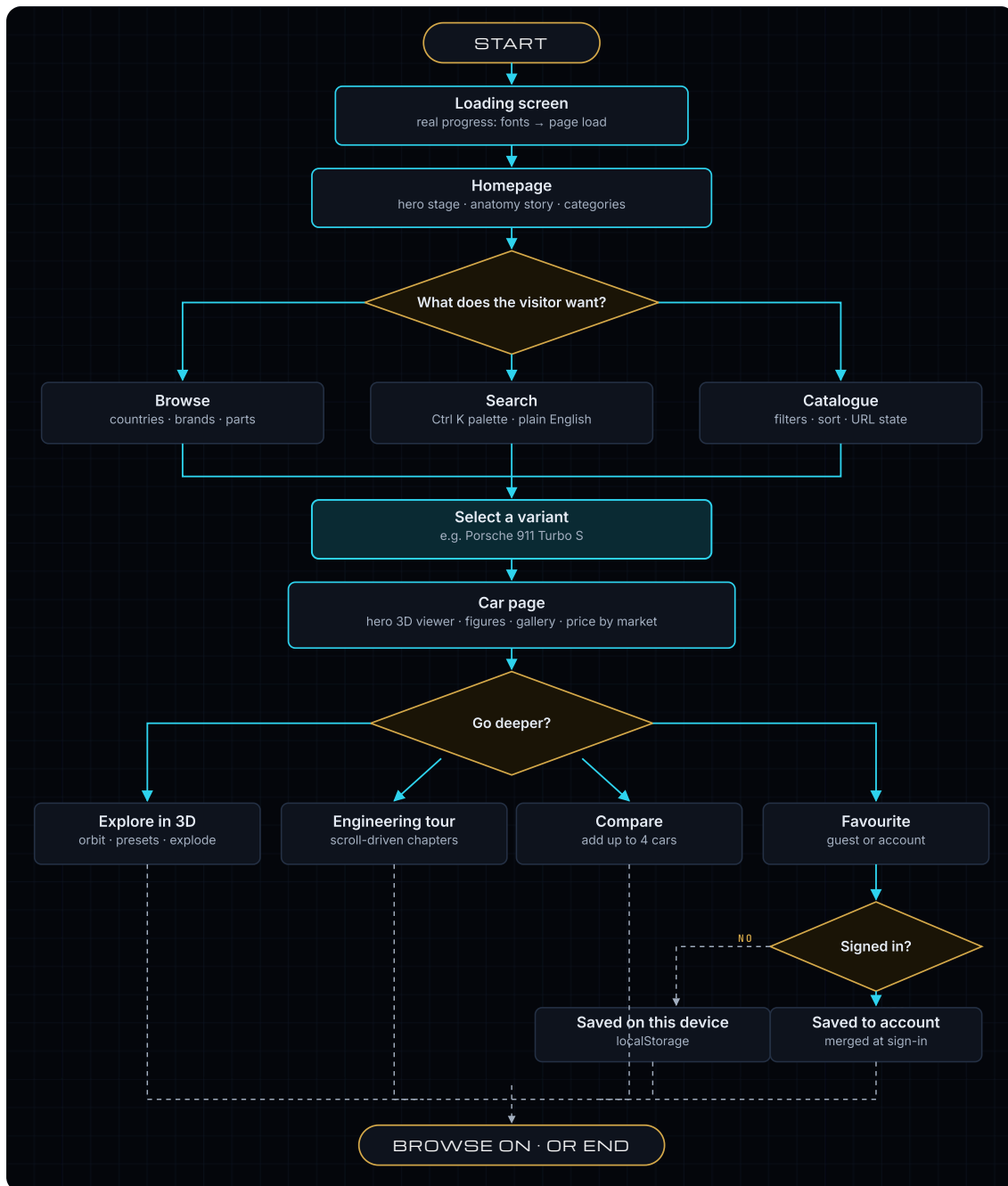


FIG 7.2 End-to-end visitor process flow

7.3 Content flow — administrator

Administrators change content through the CMS. Every step is checked twice: once by the application, and again by the database through Row Level Security. A successful save refreshes the affected pages immediately, using cache tags.

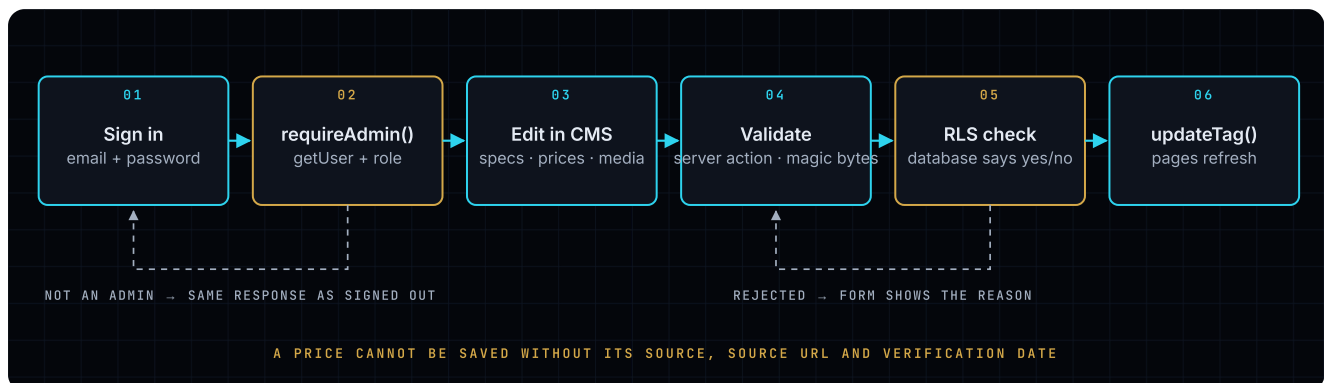


FIG 7.3 Administrator content flow, from sign-in to a refreshed public page

7.4 Rendering decision — which sections a car page shows

Spec sections are chosen from `fuel_type`, not from which rows happen to exist. The same rule decides which 3D subsystems are drawn.

FUEL TYPE	SPEC SECTIONS SHOWN	3D GROUPS
Petrol / diesel	Engine · Fuel & efficiency	Engine, exhaust — no battery
Electric	Electric drivetrain · Range & charging — no Engine section	Battery and motors — no engine
Hybrid / plug-in	Engine · Electric assist · Fuel · Electric range & charging	Engine and battery

Verified: the Tesla Model S Plaid page renders no Engine heading. The Porsche 911 Turbo S renders one engine section and no electric section. The Ferrari 296 GTB renders both, with the electric block labelled "Assist".



SECTION 08

INPUT

What visitors and administrators give the system, and what it does with it.

INPUT TYPE	DESCRIPTION	EXAMPLE
Search query	Plain-English text, parsed by ordered rules into filters; whatever is left becomes a full-text term.	german supercars above 300 km/h
Filter selections	Country, brand, body, fuel, drive, transmission, power, top speed and year. Stored in the URL, so filters can be shared.	/cars?fuel=electric
Navigation	Choosing a country, brand, model or variant.	Germany → Porsche → 911 → Turbo S
Comparison	Up to four cars; the choice lives in the URL.	?car=porsche/911/turbo-s
3D controls	Drag to orbit, scroll to zoom, preset views, explode, lighting setups, paint from sourced colours.	Tap "Engine" view
Scroll position	Drives the engineering tour and blueprint chapters; scrolling back reverses them.	Scroll through Chapter 03
Market	Country, state or city for prices; remembered on the device.	?market=india/maharashtra/mumbai
Authentication	Sign up, sign in, password reset and display name.	Email and password
Favourites	Heart toggle on any car; works for guests and accounts.	Save 911 Turbo S
Admin data entry	Vehicles, specifications, models, colours, markets and media, validated by server actions.	New variant with engine data
Price entry / CSV import	Price type, amount, currency, market, effective date, source, source URL and verification date.	Ex-showroom price with source
Media upload	Images and GLB models, validated by file signature, with licence, author and source.	CC BY-SA photograph

09

SECTION 09

SCREENSHOT OF OUTPUT

The running application. Every image in this section was captured from the production build of AURIX.

9.1 Homepage

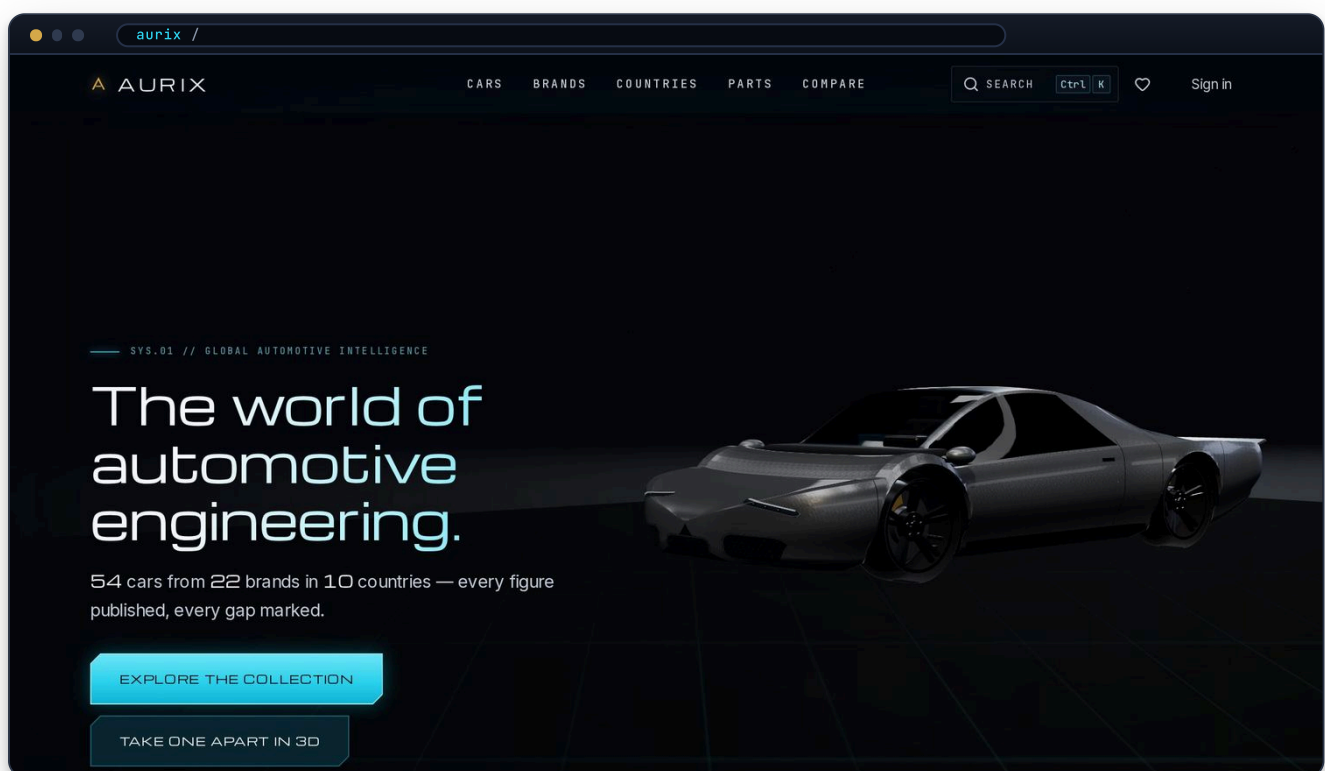


FIG 9.1 Homepage hero. The 3D stage renders the flagship car; on low-power devices a drawing to the car's published dimensions is shown instead.

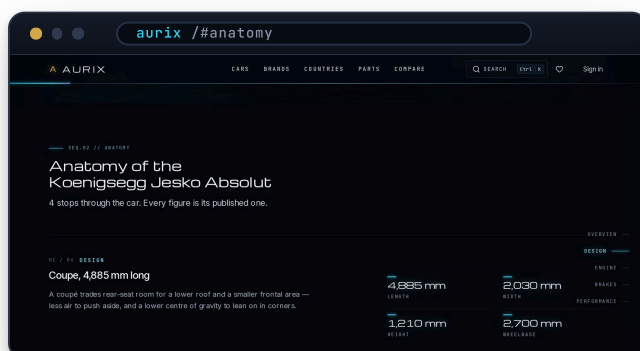


FIG 9.2 Anatomy story — the flagship car taken apart, chapter by chapter.

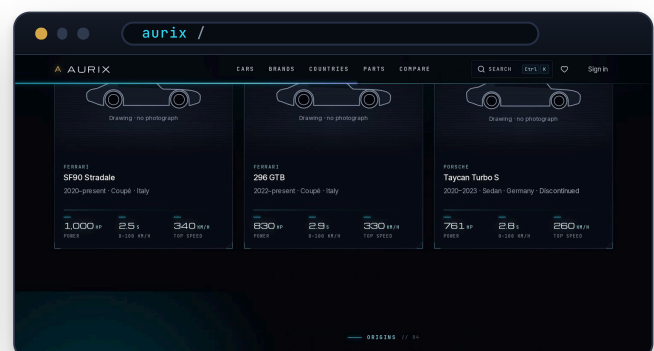


FIG 9.3 Featured cars and the origins section.

9.2 Catalogue and home sections

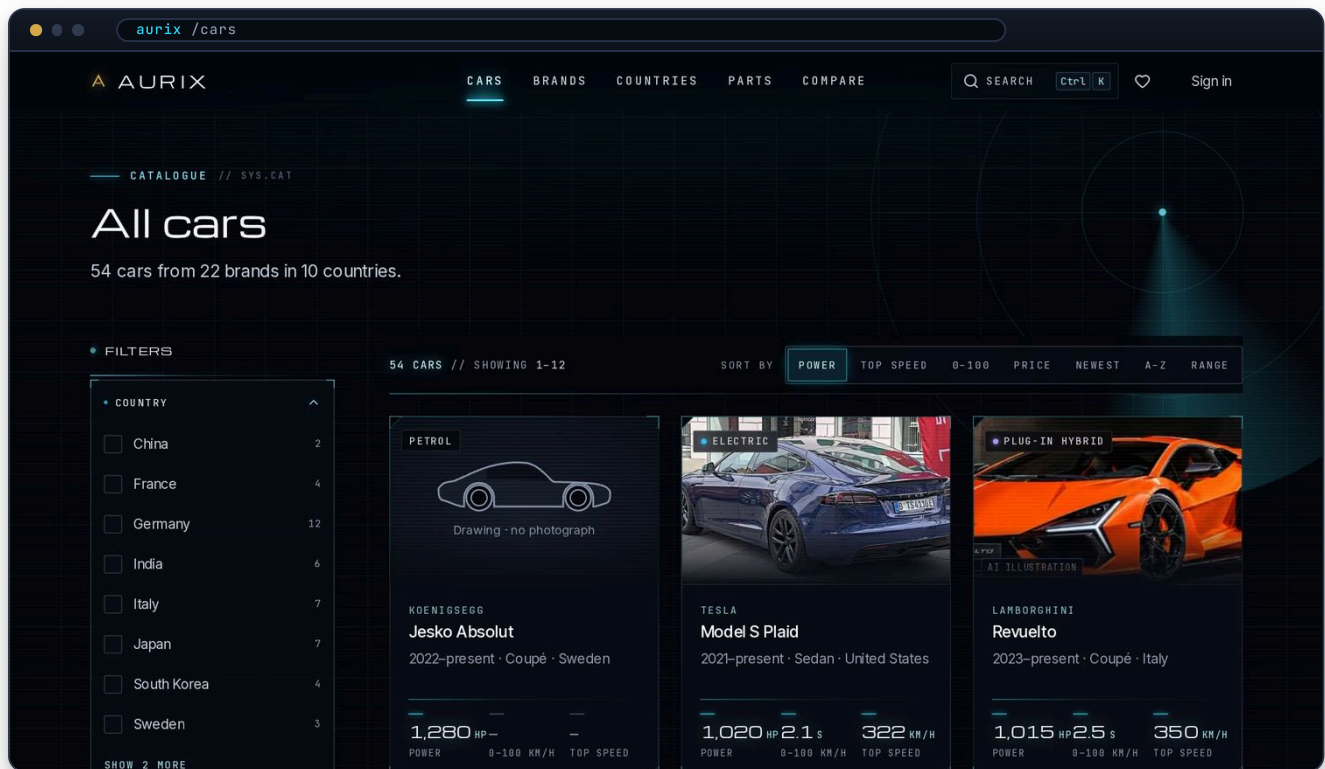


FIG 9.4 All cars. Filter rail, sort options and cards with three headline figures. Cars without a verified photograph show a body-style silhouette, labelled as a placeholder.

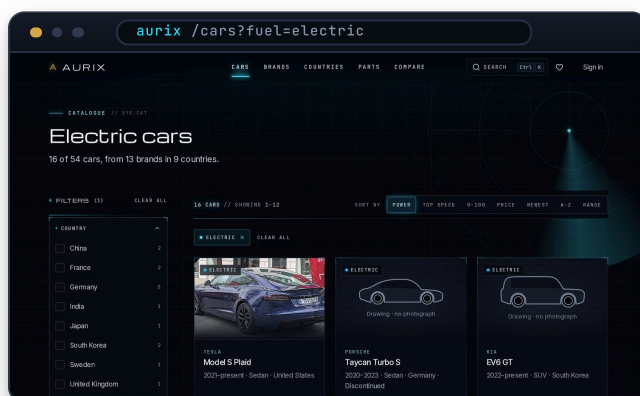


FIG 9.5 Electric cars — the heading follows the filter.

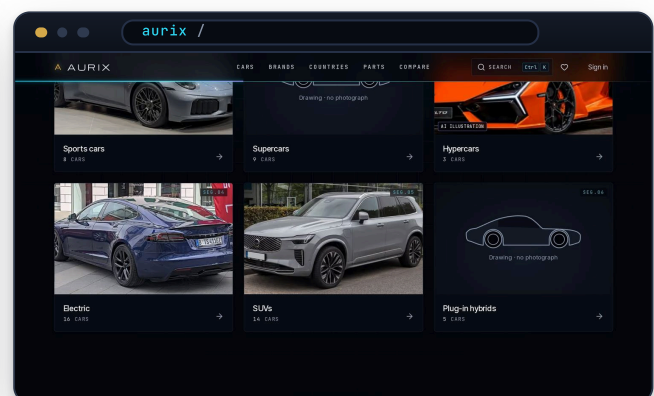


FIG 9.6 Categories with real photographs.

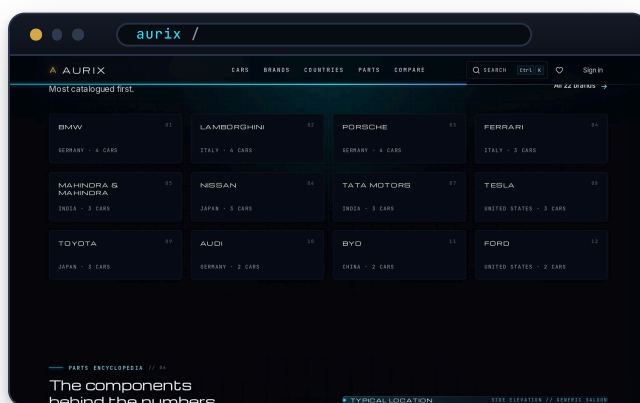


FIG 9.7 Brands and the parts encyclopedia entry.

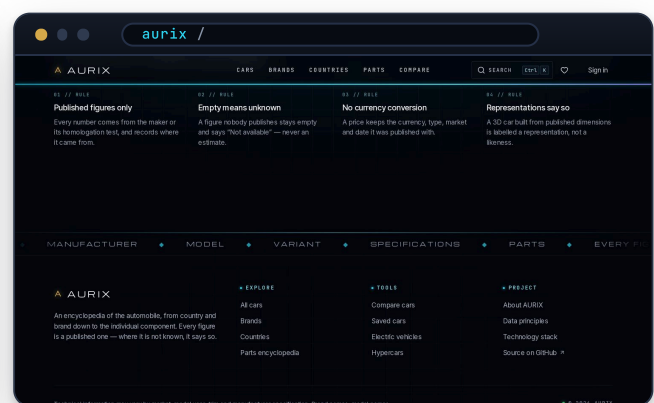


FIG 9.8 Data principles printed in the footer.

9.3 Car detail page

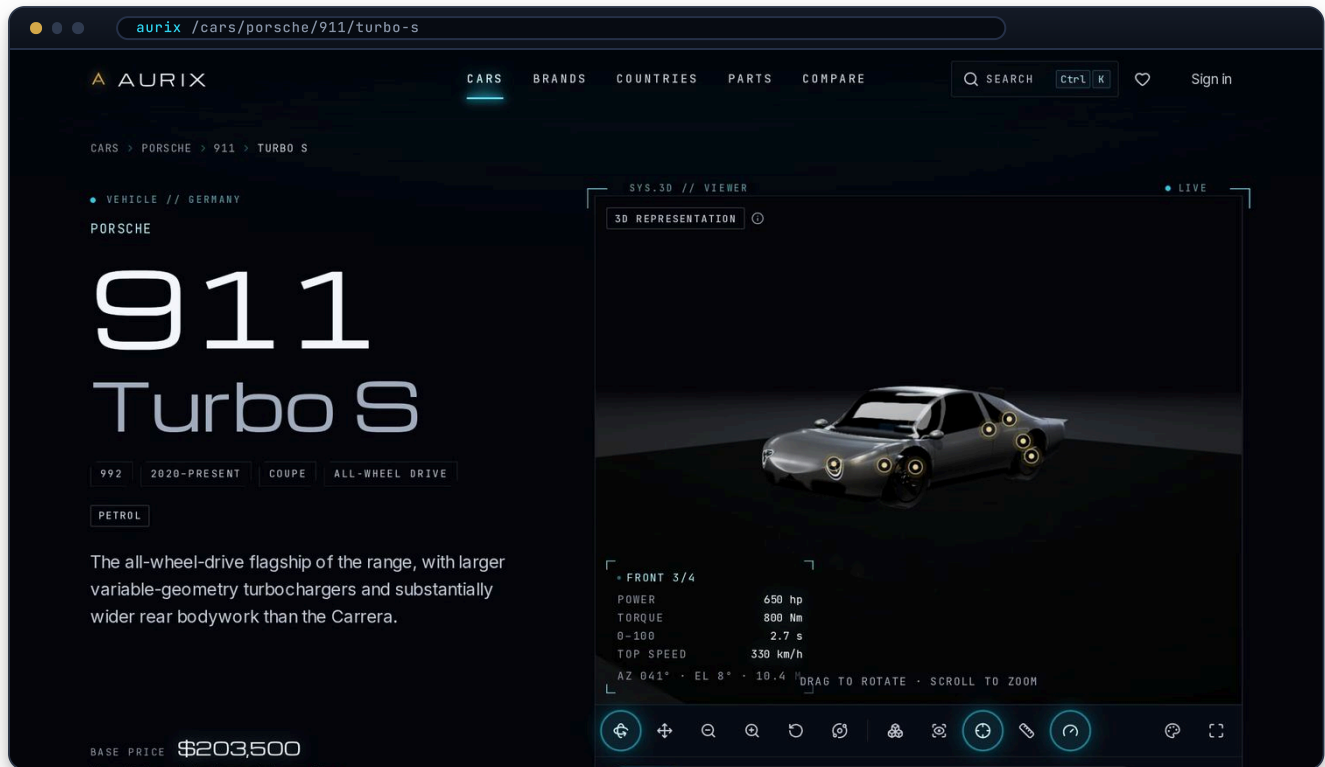


FIG 9.9 Porsche 911 Turbo S. Identity, key figures and the live 3D viewer with camera presets, lighting setups and quality control.

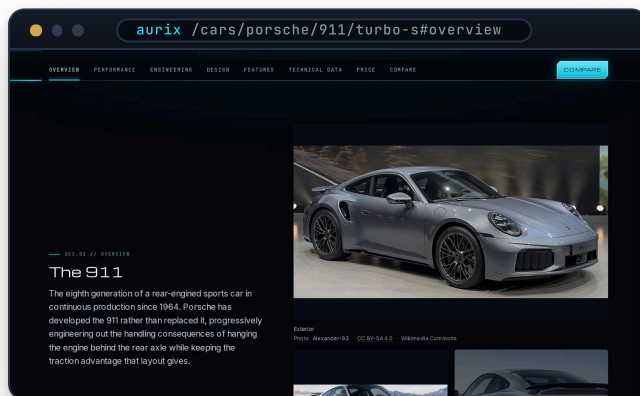


FIG 9.10 Overview with credited photographs.

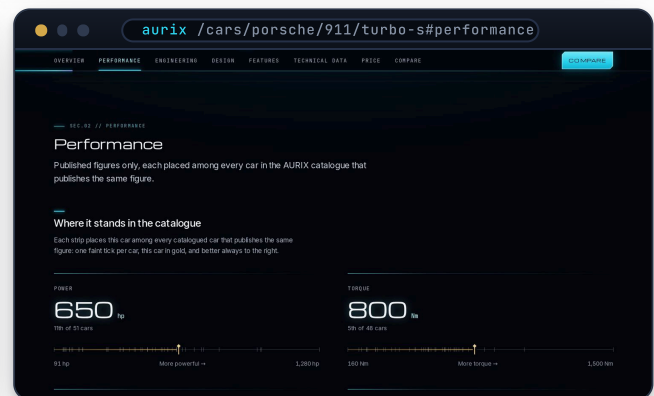


FIG 9.11 Performance — each figure ranked in the catalogue.

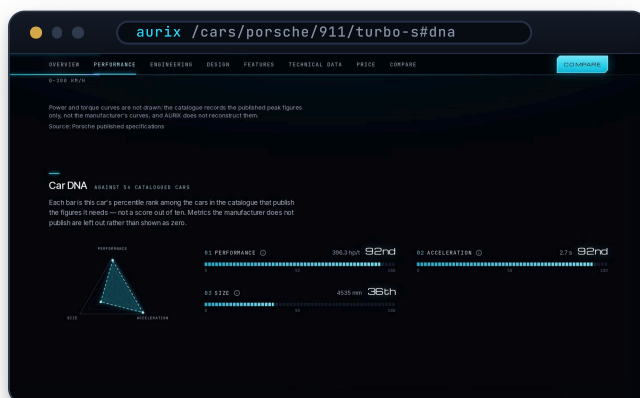


FIG 9.12 Car DNA — percentiles from real figures.

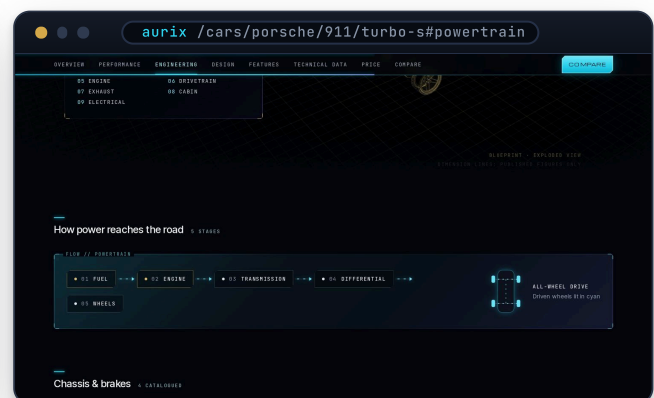


FIG 9.13 Powertrain — energy flow by fuel type.

9.4 3D blueprint and design

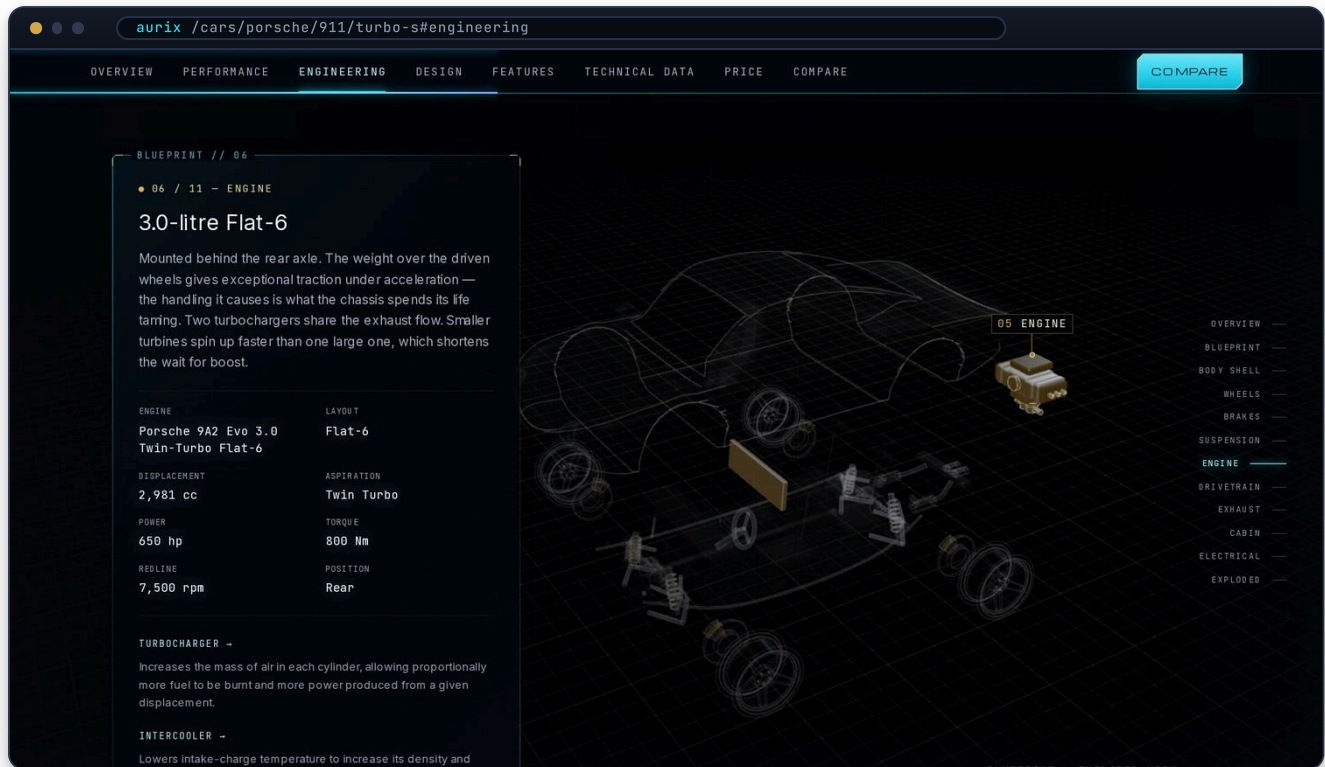


FIG 9.14 3D blueprint tour. Scrolling moves each system out of the car; the card shows only this variant's published figures (here the flat-six engine).



FIG 9.15 Body shell drawn as a gold-lined ghost.



FIG 9.16 Exploded finale — nine systems laid out.

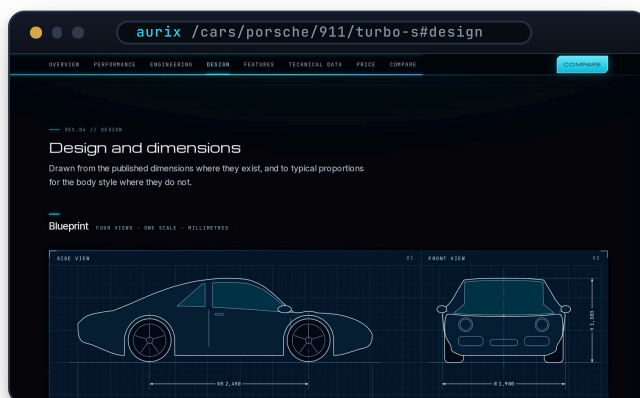


FIG 9.17 Blueprint sheet — four views, one scale.

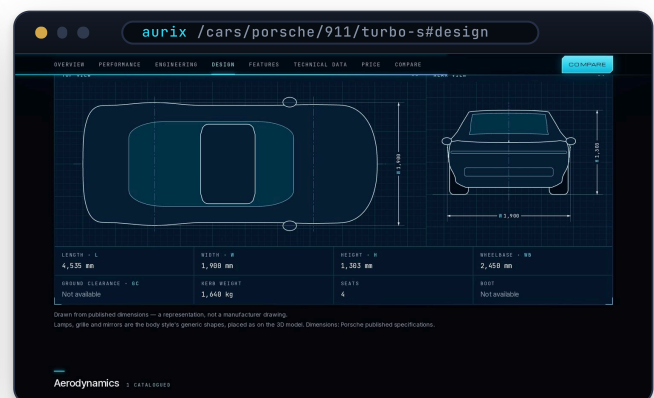


FIG 9.18 Dimensions — unpublished ones say so.

9.5 Features, technical data, pricing and EVs

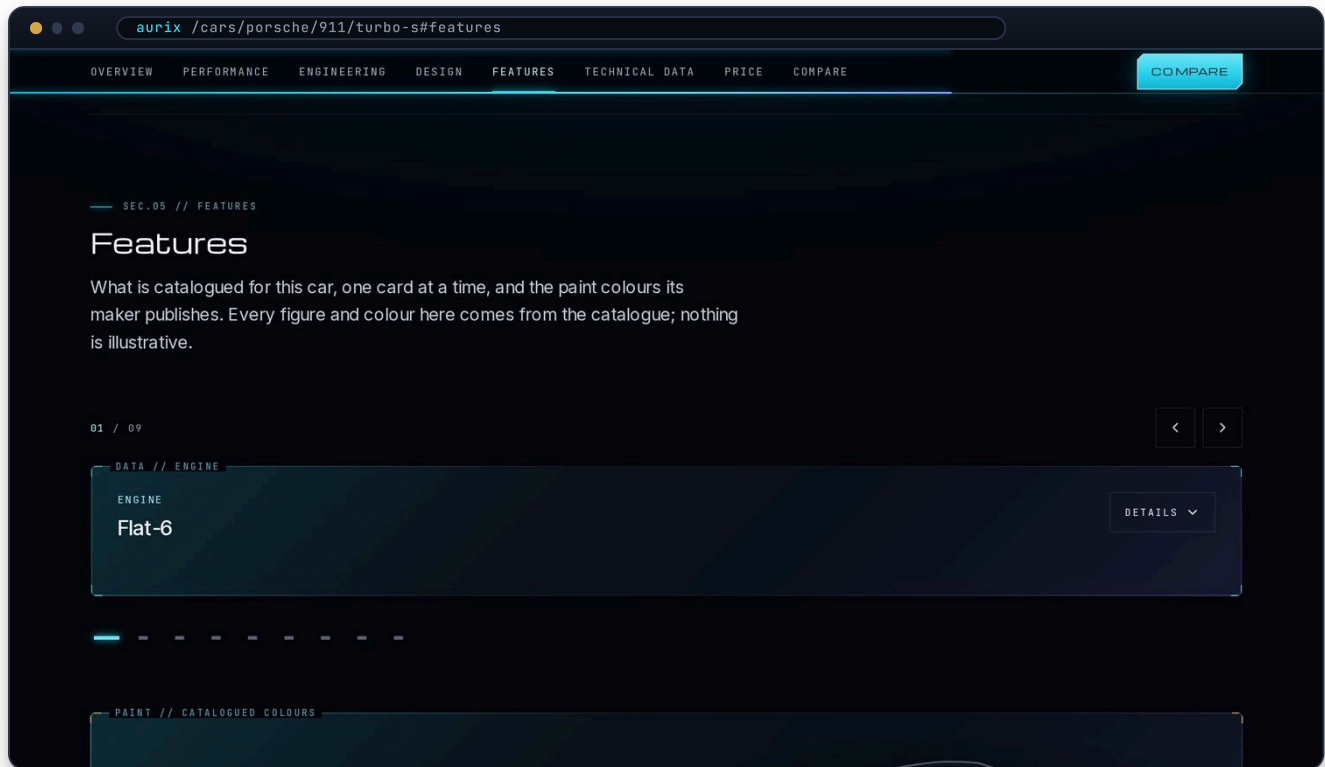


FIG 9.19 Feature showcase. A carousel of catalogued features and published subsystems that expand in place; paint colours appear only when sourced.

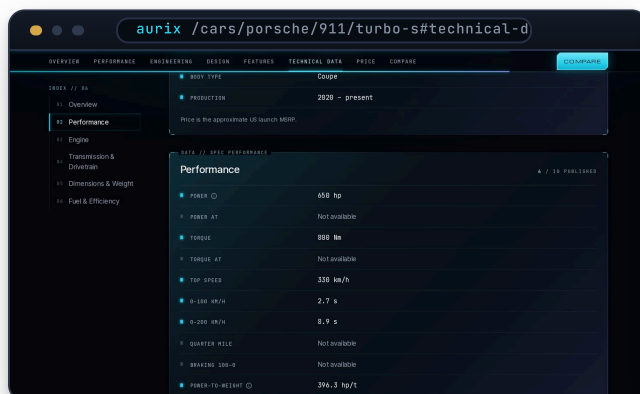


FIG 9.20 Technical data — gaps read “Not available”.

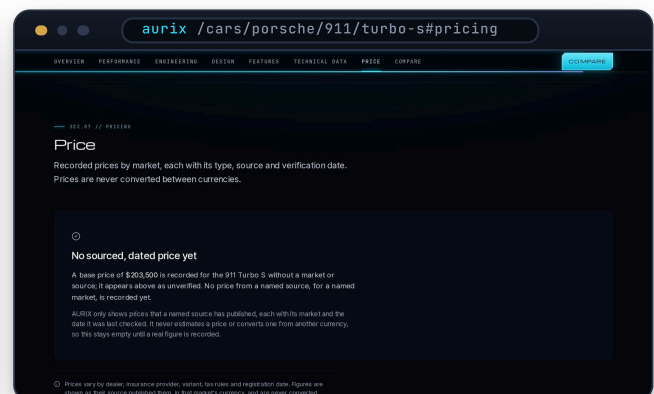


FIG 9.21 Pricing — no sourced price, so it says so.

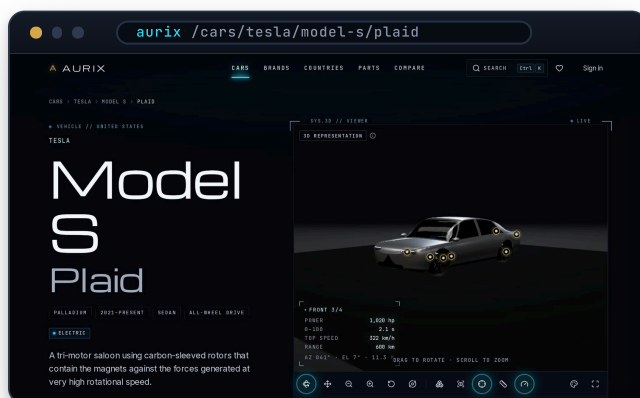


FIG 9.22 Model S Plaid — three motors, no engine.

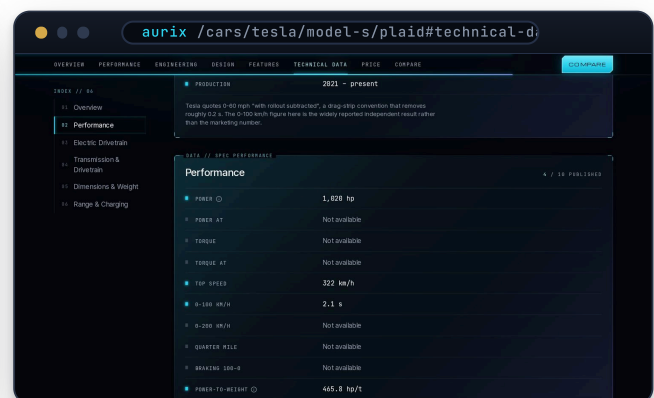


FIG 9.23 EV data — no Engine section at all.

9.6 Comparison, search and media labelling

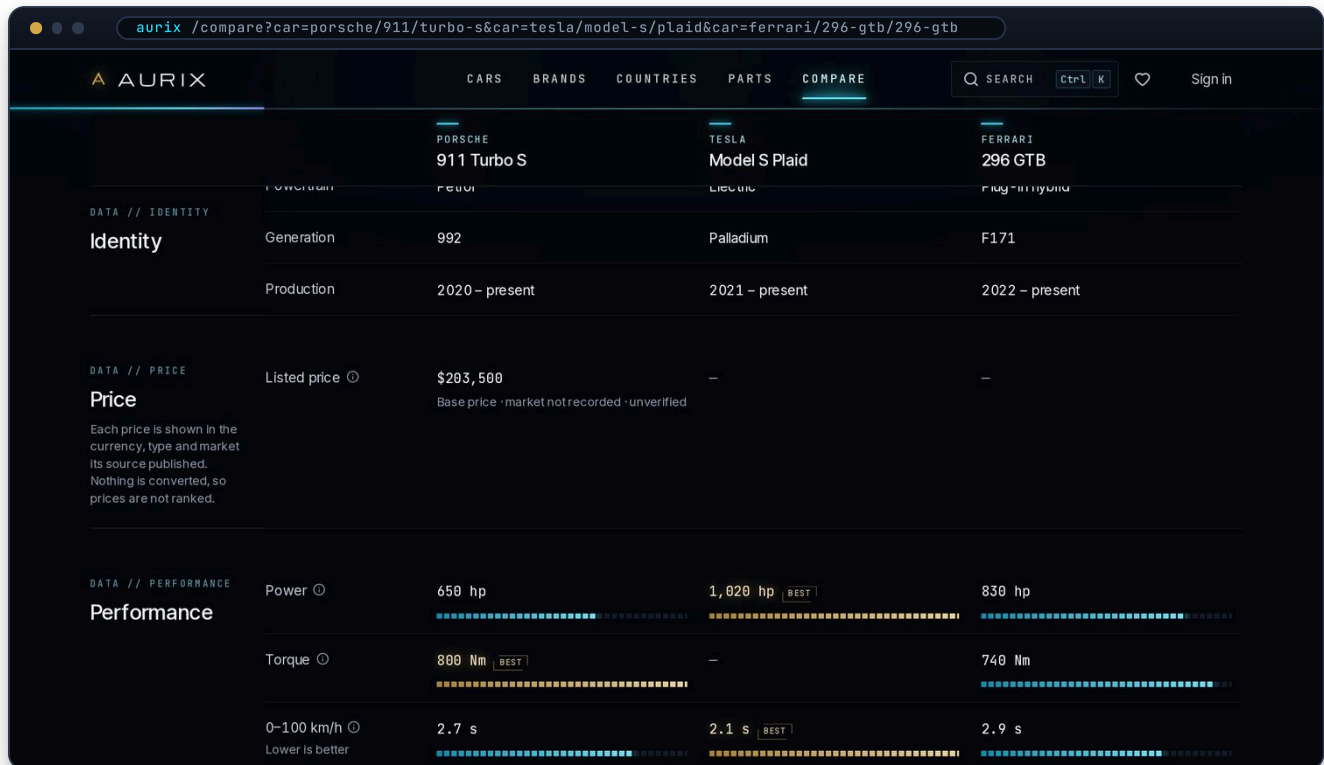


FIG 9.24 Compare. Bars only on headline performance rows; a “best” marker needs at least two published figures; prices are never ranked.

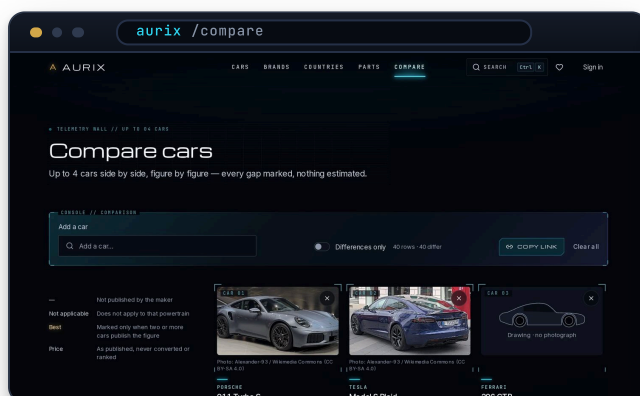


FIG 9.25 Compare — picker, legend and copy link.

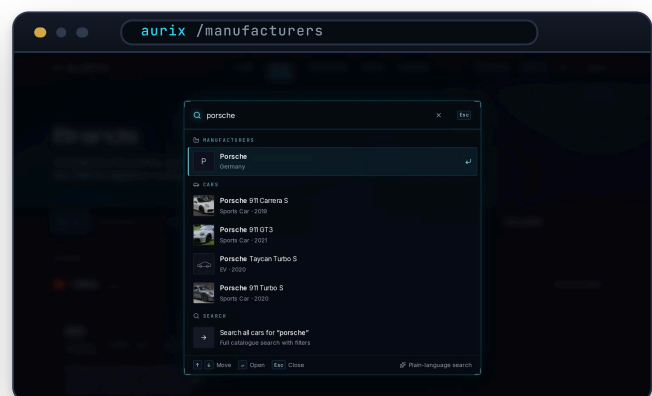


FIG 9.26 Command palette (Ctrl K) search.

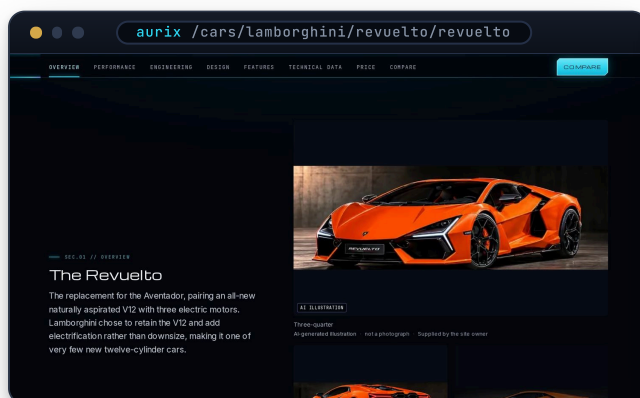


FIG 9.27 AI illustrations are badged, never “photos”.

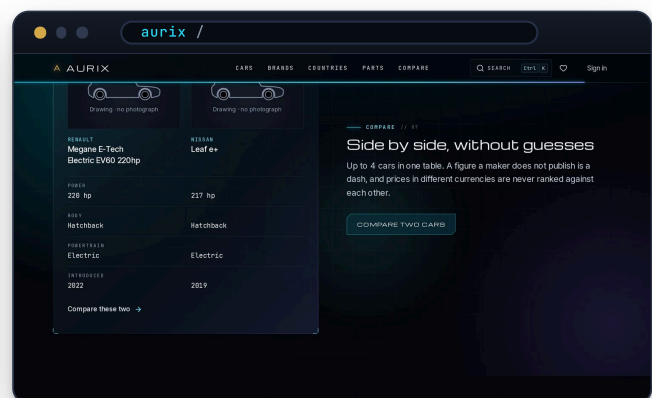


FIG 9.28 Compare entry on the homepage.

9.7 Brands, countries and parts

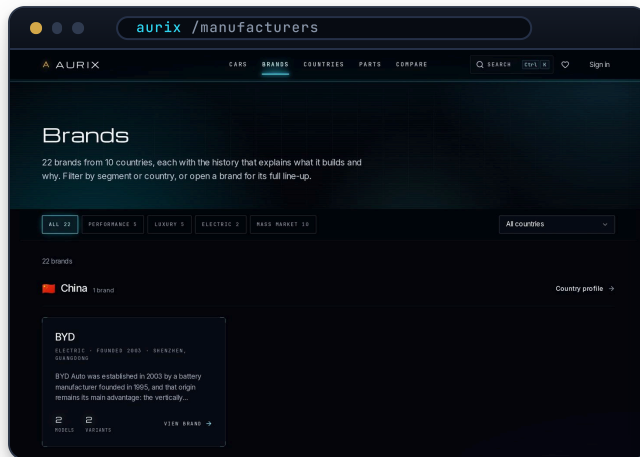


FIG 9.29 Brands — 22 brands from 10 countries.

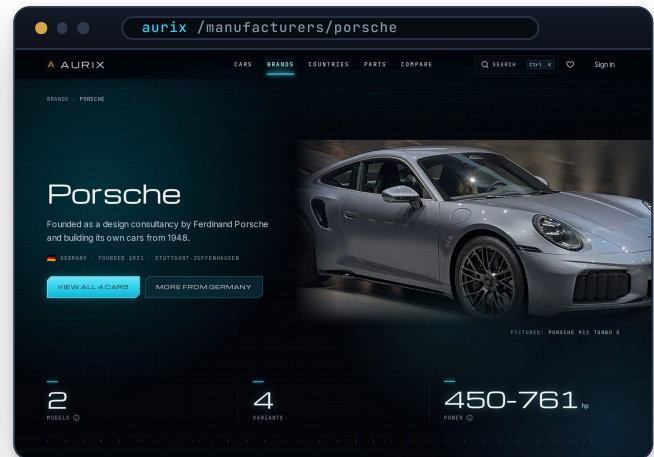


FIG 9.30 Porsche — brand profile and models.

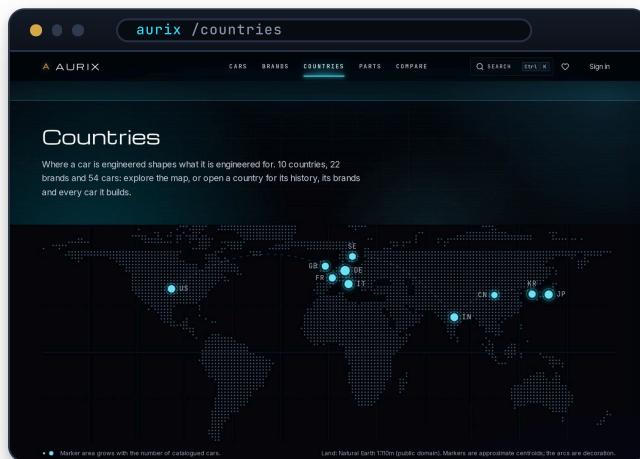


FIG 9.31 Countries — the world map explorer.

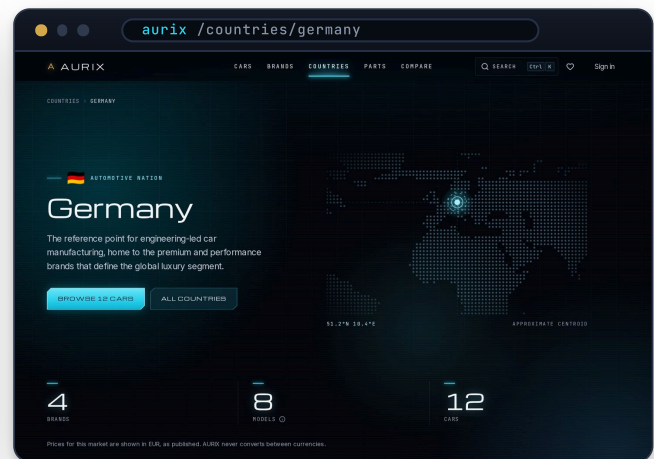


FIG 9.32 Germany — history and manufacturers.

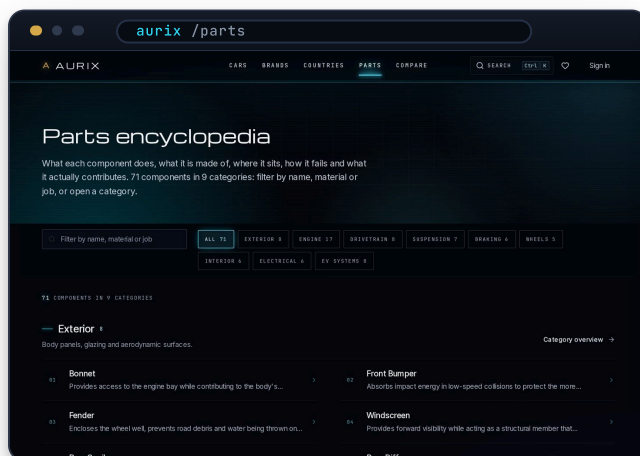


FIG 9.33 Parts encyclopedia — 71 parts in 9 systems.

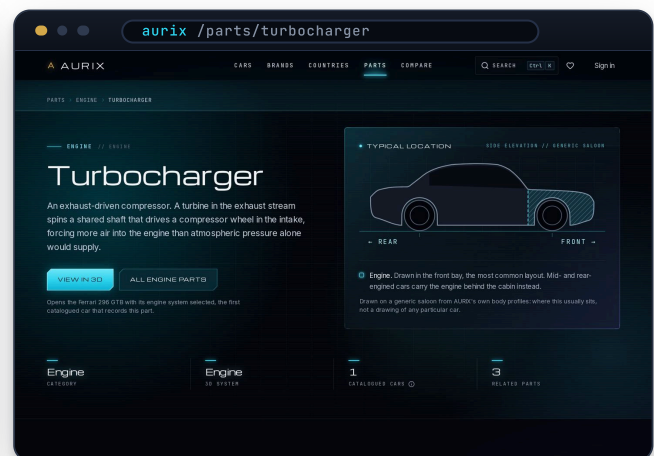


FIG 9.34 Turbocharger — what it does, which cars use it.

9.8 Admin CMS

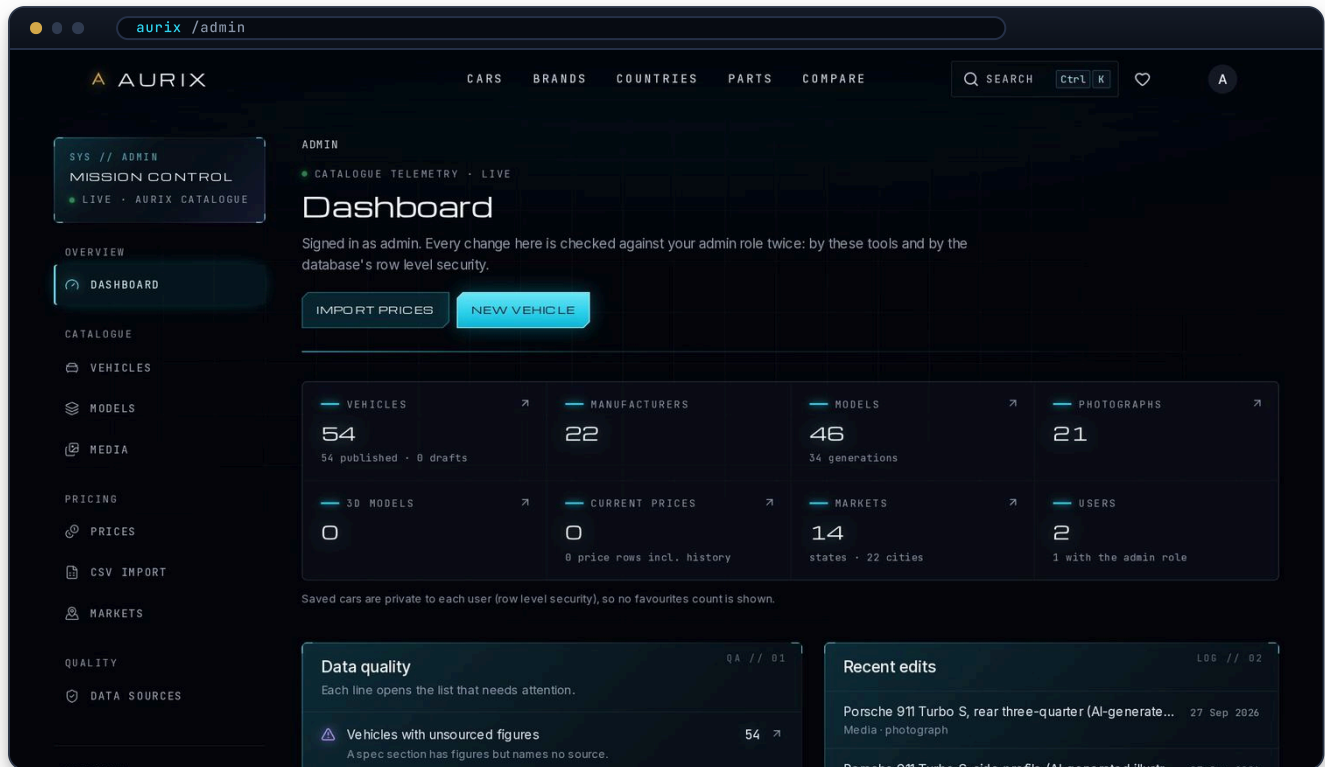


FIG 9.35 Admin dashboard. Only a signed-in user with the admin role gets here; anyone else sees the same response as when signed out.

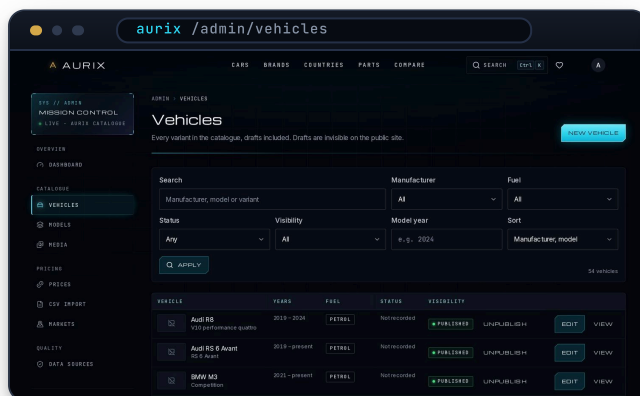


FIG 9.36 Vehicles — status and visibility, drafts included.

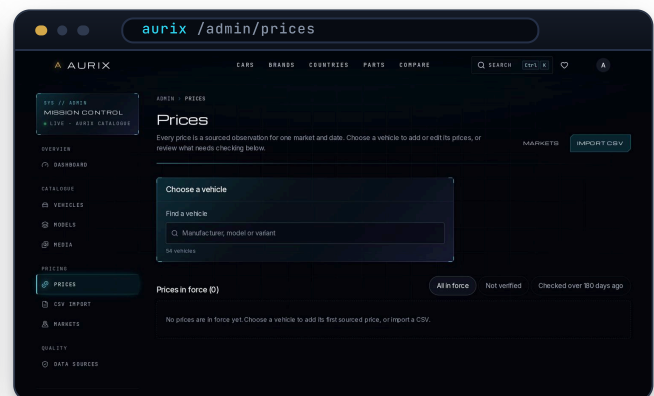


FIG 9.37 Prices — sourced entries and CSV import.

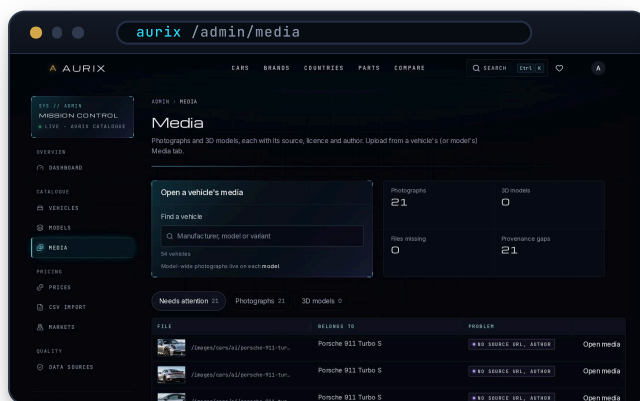


FIG 9.38 Media — files with provenance gaps flagged.

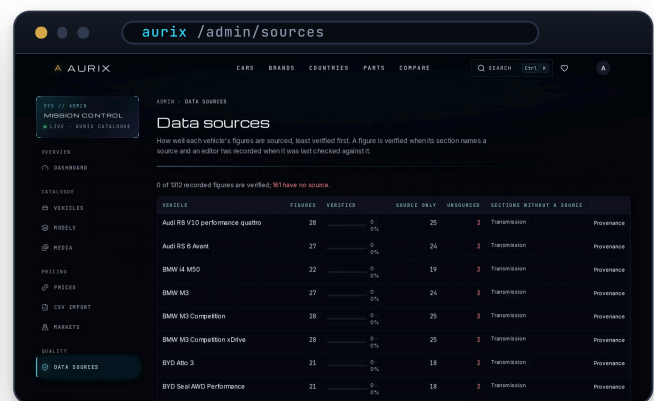


FIG 9.39 Data sources — how verified each car is.

9.9 Accounts and custom inputs

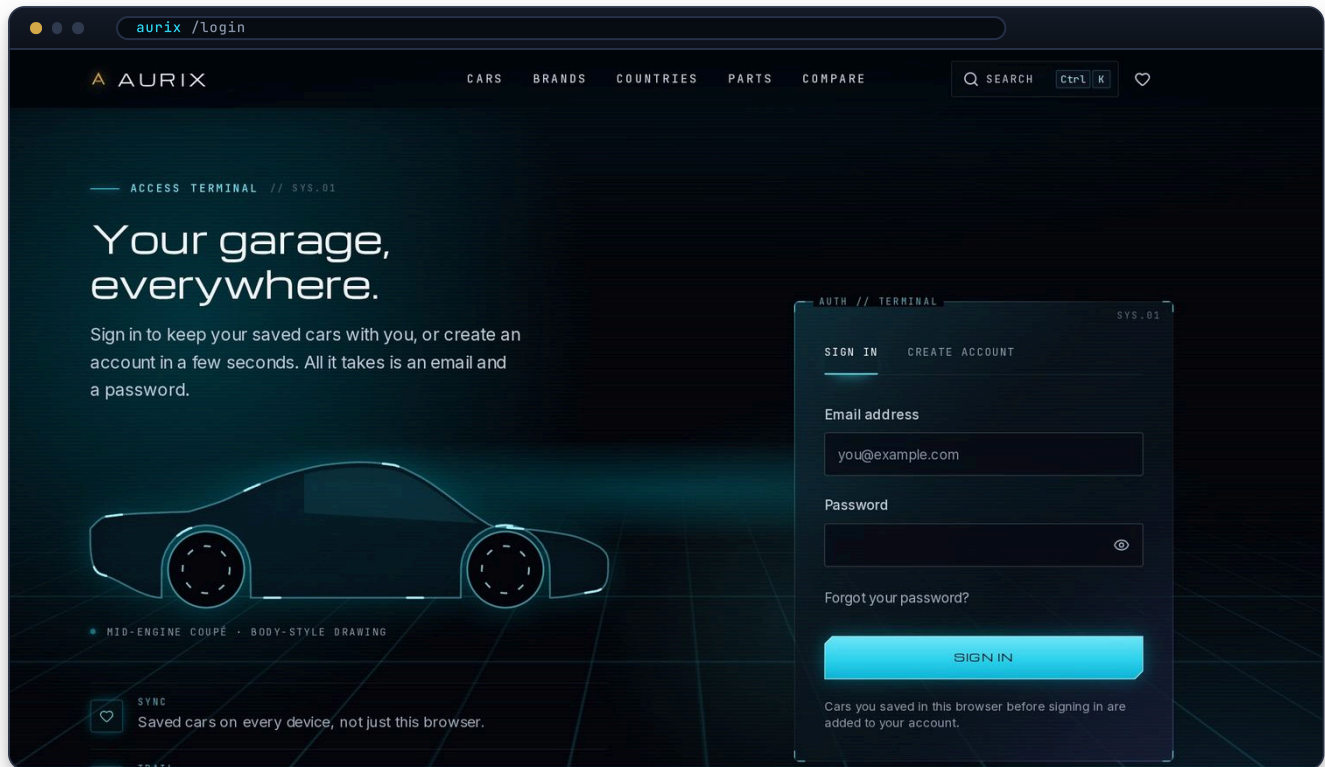


FIG 9.40 Sign in. Built on server actions, so it works without JavaScript; one generic error message for any failed sign-in.

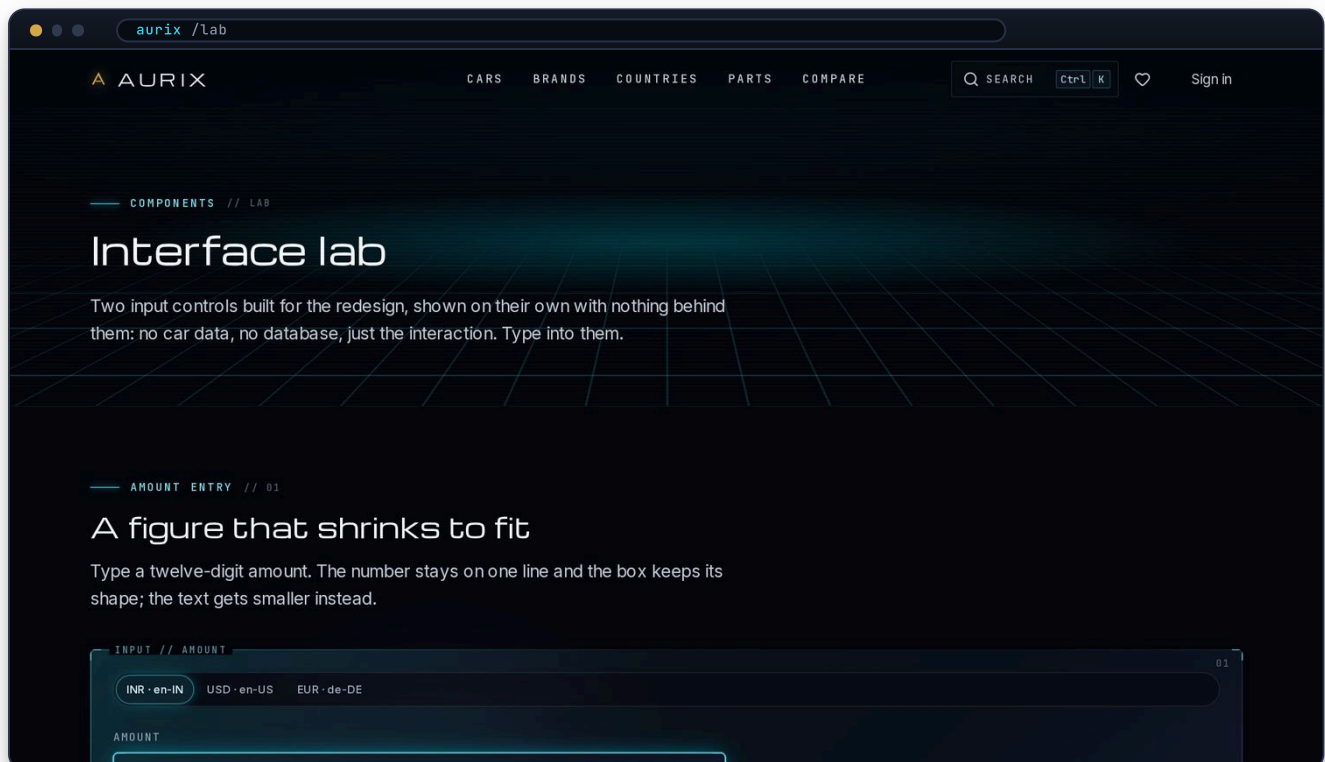


FIG 9.41 Input lab. The auto-scaling number input (Indian digit grouping shown) used for admin price amounts, and the animated-caret input used by search and sign-in.

9.10 Responsive — the same site on a phone

Every page was checked at widths from 375 to 1920 pixels with no horizontal overflow. On phones the navigation becomes a menu sheet, filters move into a bottom sheet, and the 3D viewer uses the lighter recipe.



FIG 9.42
Homepage

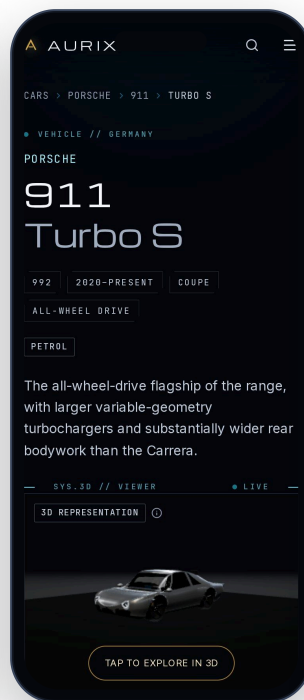


FIG 9.43
Car page

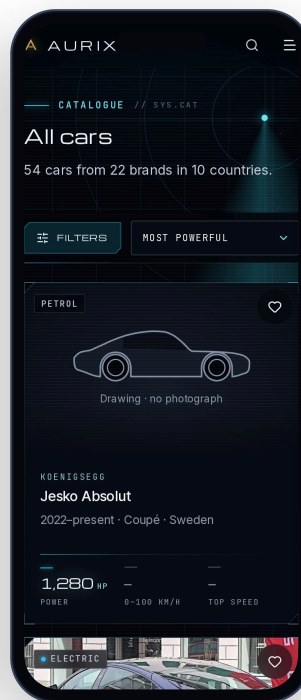


FIG 9.44
Catalogue

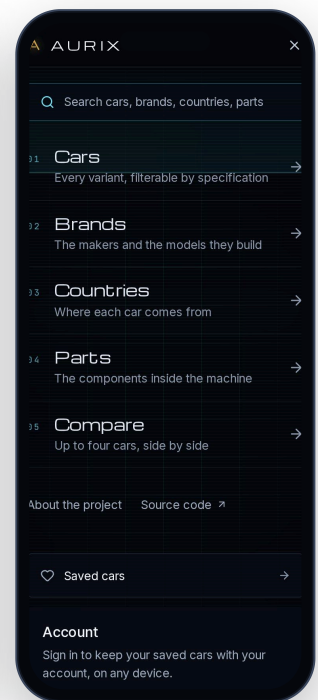


FIG 9.45
Menu sheet



FIG 9.46
Performance

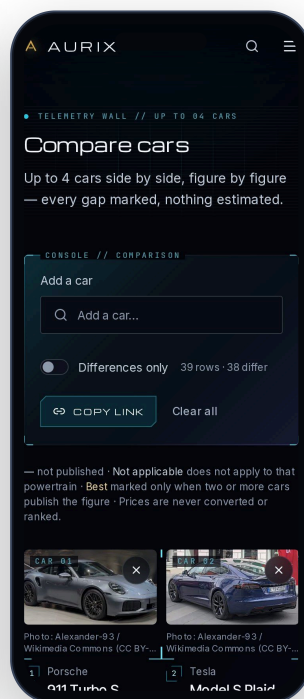


FIG 9.47
Compare

Accessibility: each page has exactly one h1, labelled inputs, a skip link, visible focus and aria-current on the active navigation item. Body text passes WCAG AA contrast on the dark background. Every 3D interaction also has a real button.

10

SECTION 10

CONCLUSION

One platform for global automotive data, interactive 3D and structured engineering information.

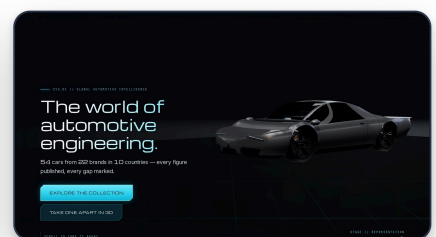
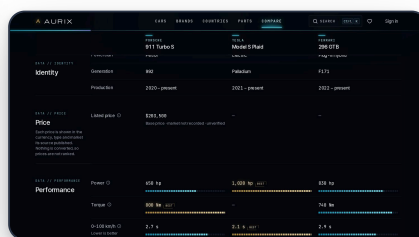
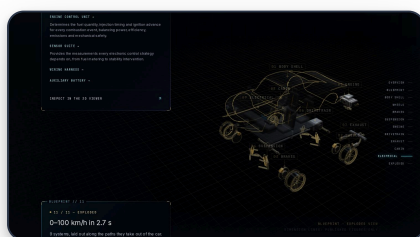
AURIX shows how a modern web stack — Next.js, React, TypeScript, React Three Fiber and Supabase — can combine global automotive data, interactive 3D and structured engineering information in one relationally sound product.

The project went beyond its original design. The planned schema became a 27-table PostgreSQL database with Row Level Security on every table. The planned 3D viewer became a parametric car built from each variant's real dimensions and layout, with an exploded view, a scroll-driven engineering tour and a blueprint. The "future" admin layer became a complete CMS with sourced pricing, validated uploads and provenance tracking. 858 automated tests and a CI pipeline keep all of this correct.

Above all, the project follows one principle of data honesty. Every figure on the site is either published by its maker or clearly marked "Not available". Prices carry their source and date, and AI illustrations are labelled as illustrations.

10.1 Skills demonstrated

- Full-stack development with server components and server actions
- Relational database design, triggers and Row Level Security
- Real-time 3D graphics and scroll-driven animation
- Responsive, accessible UI/UX and a design system
- Secure authentication and role-based administration
- Automated testing, CI and reproducible builds



CLOSING NOTE

With its modular, admin-ready structure, AURIX is ready to grow into a fully functional, continuously updated automotive intelligence platform.

SOURCES

REFERENCES

[01] Next.js 16 Documentation<https://nextjs.org/docs>**[02] React Documentation**<https://react.dev>**[03] React Three Fiber Documentation**<https://r3f.docs.pmnd.rs>**[04] drei — helpers for React Three Fiber**<https://github.com/pmndrs/drei>**[05] Three.js Documentation**<https://threejs.org/docs>**[06] GSAP and ScrollTrigger Documentation**<https://gsap.com/docs>**[07] Supabase Documentation (Auth, RLS, Storage)**<https://supabase.com/docs>**[08] PostgreSQL Documentation**<https://www.postgresql.org/docs>**[09] Tailwind CSS v4 Documentation**<https://tailwindcss.com/docs>**[10] TypeScript Handbook**<https://www.typescriptlang.org/docs>**[11] Vitest Documentation**<https://vitest.dev>**[12] Playwright Documentation**<https://playwright.dev>**[13] Web Content Accessibility Guidelines (WCAG) 2.2**<https://www.w3.org/TR/WCAG22>**[14] MDN Web Docs (HTML, CSS, JavaScript, WebGL)**<https://developer.mozilla.org>**[15] Wikimedia Commons — licensed vehicle photographs (credits in public/images/CREDITS.md)**<https://commons.wikimedia.org>**[16] Vercel Documentation**<https://vercel.com/docs>

01 BODY SHELL

AURIX

GLOBAL AUTOMOTIVE INTELLIGENCE

Every figure published.

Every gap marked.

Zenil Rana · Sairaj Bhadale · Saket Shrivastava

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